

Taylor & Francis
Taylor & Francis Group

Structure and Infrastructure Engineering
Maintenance, Management, Life-Cycle Design and Performance

ISSN: 1573-2479 (Print) 1744-8980 (Online) Journal homepage: www.tandfonline.com/journals/nsie20

Detecting, monitoring and modeling damage within the decision-making process in the context of managing bridges: a review

Ana Carolina Pereira Martins, José Maria Franco de Carvalho, Maria Cláudia Sousa Alvarenga, Diôgo Silva de Oliveira, Kléos Magalhães Lenz César Júnior, José Carlos Lopes Ribeiro, Galileu Silva Santos & Rogério Calazans Verly

To cite this article: Ana Carolina Pereira Martins, José Maria Franco de Carvalho, Maria Cláudia Sousa Alvarenga, Diôgo Silva de Oliveira, Kléos Magalhães Lenz César Júnior, José Carlos Lopes Ribeiro, Galileu Silva Santos & Rogério Calazans Verly (2026) Detecting, monitoring and modeling damage within the decision-making process in the context of managing bridges: a review, Structure and Infrastructure Engineering, 22:3, 565-587, DOI: 10.1080/15732479.2024.2331103

To link to this article: https://doi.org/10.1080/15732479.2024.2331103

Published online: 25 Mar 2024.

Submit your article to this journal

Article views: 726

View related articles

View Crossmark data

Citing articles: 13 View citing articles

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=nsie20

STRUCTURE AND INFRASTRUCTURE ENGINEERING
2026, VOL. 22, NO. 3, 565–587
https://doi.org/10.1080/15732479.2024.2331103

Taylor & Francis
Taylor & Francis Group

REVIEW ARTICLE

Check for updates

Detecting, monitoring and modeling damage within the decision-making process in the context of managing bridges: a review

Ana Carolina Pereira Martins[a], José Maria Franco de Carvalho[a], Maria Cláudia Sousa Alvarenga[a], Diôgo Silva de Oliveira[a], Kléos Magalhães Lenz César Júnior[a], José Carlos Lopes Ribeiro[a], Galileu Silva Santos[b] and Rogério Calazans Verly[b]

[a]Department of Civil Engineering, Federal University of Viçosa, Viçosa, Brazil; [b]National Department of Transportation Infrastructure (DNIT), Brasília, Brazil

ABSTRACT

The expansion of transportation infrastructure and the aging and deterioration of its constituent elements make bridge maintenance management programs more expensive and complex. In this context, a bridge management system (BMS) has become a fundamental tool for managing and controlling the entire process involving the structures, from design, construction, operation, and maintenance. Information regarding bridges, inspection, and damage detection should be standardized and digitized for stakeholder access. The provided bibliometric analysis demonstrates that inspection, structural health monitoring (SHM), deterioration, damage detection, and decision-making are trending topics. These topics guided a comprehensive literature review bringing advances and discussing assessment quality, the ability to detect damage, and the most accurate and cost-effective intervention. Finally, the challenges and limitations of these topics are identified, and possible solutions to overcome these limitations are discussed.

ARTICLE HISTORY
Received 19 July 2023
Revised 18 December 2023
Accepted 22 January 2024

KEYWORDS
bridge management system (BMS); building information modeling (BIM); damage information modeling (DIM); damage detection; inspection; maintenance management; structural health monitoring (SHM)

1. Introduction

Bridges are fundamental components of the transportation system, economic vitality, and personal mobility, facilitating access between different areas through the transposition of barriers (Kendall, Keoleian, & Helfand, 2008; Marzouk & Hisham, 2012; Wan et al., 2019). The growing and aging of the transportation infrastructure put pressure on developing more efficient technologies and tools to assess and manage these structures over their lifetime (Abudayyeh & Al-Battaineh, 2003; Boddupalli, Sadhu, Rezazadeh Azar, & Pattyson, 2019; Costin, Adibfar, Hu, & Chen, 2018; Sedek & Serwa, 2016). Due to the nature of loads and the environmental conditions in which they are inserted, the bridge elements are subject to rapid deterioration, requiring intervention actions of maintenance, rehabilitation, or replacement (Frangopol & Soliman, 2015; Mohammadi, Guralnick, & Yan, 1995; Xie, Wu, & Wang, 2018).

Since various factors affect a bridge's performance, safety, maintenance, and cost, the intervention process chosen to rehabilitate or replace some elements is a difficult task (Benítez, Rocha, Varum, & Rodrigues, 2020; Mohammadi et al., 1995). Besides an efficient and well-developed design, for a structure to perform satisfactorily throughout its service life, it is necessary to consider other durability analyses and parameters, such as the geometry of the designed elements, the materials used in the construction, traffic volume analysis, and environmental analysis (Pipinato, 2016).

Considering the difficulties encountered in the application of available bridge management methodologies, where decisions are made at a single level, in many countries, agencies responsible for infrastructure networks have made significant efforts to develop bridge management systems (BMS) that assess the structure's condition at the overall network level during its life cycle and provide information on the efficiency of resource allocation and management policy setting (Jeong, Hou, Lynch, Sohn, & Law, 2017; Pipinato, 2016; Shepard, 2005). BMS is used in data collection, inspection process, condition assessment, maintenance planning, repair or replacement, optimizing financial resources, and increasing user safety (Alonso Medina & León González, 2022; Hurt & Schrock, 2016; Marzouk & Hisham, 2012; Shepard, 2005).

Building Information Modeling (BIM) is an efficient tool for digitally representing a structure's physical and functional characteristics, comprising life cycle information and costs throughout the design, management, monitoring, and repair. It is a methodology and technology usually used to create a digital twin (DT) of a physical asset, such as a building or infrastructure. In this context, DT technology emerges, combining the real and digital worlds, such as a digital replica model of a physical object (Hosamo & Hosamo, 2022; Nguyen, Kang, et al., 2022). BIM also supports integration with the Internet of Things (IoT), sensors, 3D scanning, machine learning, big data, and artificial

CONTACT Ana Carolina Pereira Martins ana.martins5@ufv.br Department of Civil Engineering, Federal University of Viçosa, Viçosa, Brazil.
© 2024 Informa UK Limited, trading as Taylor & Francis Group

intelligence (AI) to provide real-time data and collaboration, making it an essential tool for optimizing efficiency and decision-making (Chong, Lopez, Wang, Wang, & Zhao, 2016; Costin et al., 2018). In addition, Fanning, Clevenger, Ozbek, and Mahmoud (2015) demonstrated the economic benefits of using BIM in bridge construction, potentially reducing costs by approximately 5–9% with reduced waste, increased efficiency, and sustainability of projects throughout the life cycle.

BIM-based standardization through parametric modeling offers possibilities for improving the quality and efficiency of facility management operations to create, share, exchange, and manage information among all stakeholders, facilitating integration, interoperability, and collaboration (Chen & Wang, 2009; Isikdag, Underwood, & Aouad, 2008). The Industry Foundation Classes (IFC) schema represents the BIM database. This data model is an open standard developed by BuildingSMART and is widely used for data exchange (ISO, 2018). IFC is an object-oriented data architecture that allows the structuring of geometric data and project information in a standardized way (buildingSMART, 2023). Until version 4, the IFC was focused on buildings, but an extension project was initiated with the growing international demand for a data schema for infrastructure (Byun, Han, Kwon, & Kang, 2021).

This study provides a comprehensive and updated review of techniques and technologies of monitoring, damage detection, damage information modeling (DIM), and decision-making processes within a BMS. This article's novelty is a literature review that covers all the issues surrounding the aging and deterioration of the enormous number of bridges worldwide and the high cost associated with maintenance, reinforcement, and rehabilitation of these structures. It emphasizes the role of inspection and structural health monitoring (SHM) in the timely detection of failures and damage, assisting in decision-making on maintenance and repair actions. Finally, the challenges and limitations of these topics are identified and discussed, with the proposition of possible solutions to overcome these limitations.

This article is organized as follows. First, Section 2 covers the technical background: 2.1 Bridge Maintenance Management and 2.2 Building Information Modeling (BIM) in a Bridge Management System (BMS). Section 3 presents the method applied in the literature review. Section 4 addresses the scientific interest and trends in bridge management. Section 5 incorporates the literature review and remarks about some specific themes: 5.1 Monitoring, 5.1.1 Inspection, 5.1.2 Structural health monitoring (SHM), 5.2 Deterioration and damage, 5.2.1 Damage detection, 5.2.2 Damage information modeling (DIM), 5.3 Decision-making process, and 5.4 Challenges and limitations. Section 6 summarizes the concluding remarks of the study.

2. Background

2.1. Bridge maintenance management

Bridge maintenance management plays a vital role in extending the service life and ensuring the safe operation of structures (Costin et al., 2018; Gui et al., 2021; Yang et al., 2022). A well-planned maintenance strategy should focus on enhancing the operational process, reducing the frequency of maintenance actions, addressing complexity, and streamlining the skills required for maintenance (Benítez et al., 2020). The maintenance management system comprises several key components (Figure 1), including information collection through inspections and monitoring, damage identification through technical assessments, maintenance decision-making based on inspection reports, and the actual implementation of maintenance through repair, rehabilitation, and reconstruction (Gui et al., 2021; Ren, Ding, & Li, 2019; Wu et al., 2021; Yin, Li, Guo, & Li, 2011).

Maintenance operations can be classified into three main groups (Figure 2): preventive, predictive, and corrective, and they aim to ensure the structure's integrity, preserving it from deterioration. Preventive maintenance consists of periodic actions carried out in advance to avoid system failures and keep the structure performing above a minimum performance level (Frangopol & Kim, 2011; P. Helene, 2005; Sánchez-Silva, Frangopol, Padgett, & Soliman, 2016; Xie et al., 2018). Preventive and routine maintenance represent actions necessary to preserve the structural and operational efficiency of the bridge, including activities that maintain the serviceability of the structure before it reaches a level of deterioration that requires corrective maintenance (Benítez et al., 2020; PennDOT, 2012; Rashidi, 2013). In general, routine actions do not change the bridge's condition and include cleaning the drainage, minor repairs to the road surface and deck joints, and removing debris (NYSDOT, 2008; Rashidi, 2013).

Essential or predictive maintenance is associated with performance-based monitoring and is promptly executed when specific indicators reach predefined target values, with particular emphasis on those related to the structural condition (Flores-Colen & De Brito, 2010; Frangopol & Kim, 2011; Xie et al., 2018). However, this can be highly debatable in practice, as the optimal solution often requires striking a balance among different indicators, even when pushing some of them to extremes (e.g. long-term costs).

Both preventive and predictive maintenance tend to focus on the technical condition of bridges (Gui et al., 2021). Intervention actions carried out when the minimum performance limit is achieved or exceeded are called corrective or reactive maintenance (Akcamete, Akinci, & Garrett, 2010; Benítez et al., 2020; Sánchez-Silva et al., 2016). Much of the maintenance work is reactive, and this practice is ineffective since corrective maintenance tasks can cost three to four times more than the same repair activity performed planned (Akcamete et al., 2010; Flores-Colen & De Brito, 2010).

Proper maintenance planning can optimize costs during repair interventions in the bridge life cycle (Nili, Taghaddos, & Zahraie, 2021). The selection of the start time and time interval of preventive maintenance will influence both life cycle cost and the environmental impact of existing bridges (Xie et al., 2018). The extent of servicing also plays a relevant role in defining an optimal maintenance program, as it

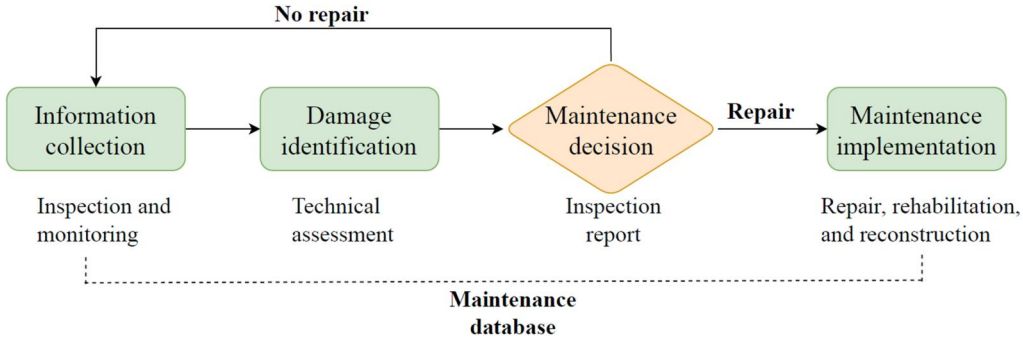


Figure 1. Bridge maintenance information and decision structure.

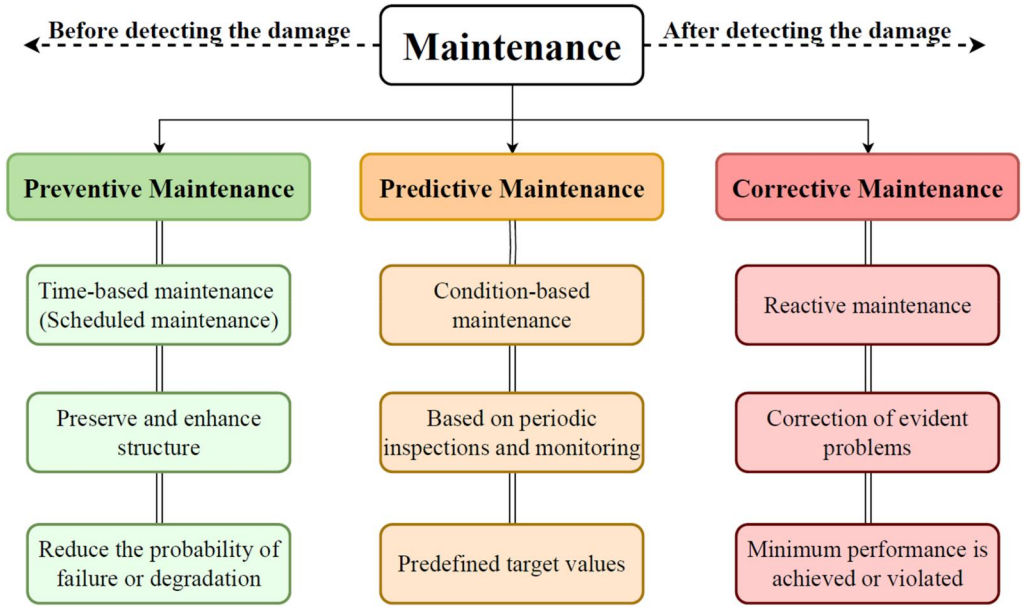


Figure 2. Main types of maintenance and their characteristics.

sets the costs and influences the availability of services (Sánchez-Silva et al., 2016).

Many bridges have been built over the past decades around the world, especially after the Second World War in the late 1940s. These structures were not designed for nowadays traffic following outdated standards, and need to be rehabilitated after a certain period of operation to ensure their safety and serviceability (Mohammadi et al., 1995; Rashidi et al., 2020; Xie et al., 2018). Over its entire life cycle, a bridge consumes between 0.4 and 2.0% of its construction cost per year in operation, inspection, maintenance, and demolition actions, corresponding to up to 80% of the construction cost (Artus & Koch, 2020).

The average age of the 617,000 bridges in the United States is 44 years, with 42% being at least 50 years old (ASCE, 2021). The annual budget of approximately $14.4 billion is spent on the maintenance of existing bridges, and this budget needs to be increased to $22.7 billion annually or by 58% to improve the condition of the structures since 7.5% of the country's bridges are considered structurally deficient (ASCE, 2021).

Japan has approximately 730,000 road bridges and 11,000 road tunnels across the country. It is estimated that about 39% of bridges and 27% of tunnels will be over 50 years old in 2023 (MLIT, (Ministry of Land, Infrastructure, Transport & Tourism, & Japan), 2019). China has also experienced increased demand for the maintenance and management of its bridges due to infrastructure development and expansion (Wan et al., 2019). There are more than 800,000 highway bridges in China, and more than 1/3 of these bridges have structural failures in various degrees of damage, requiring safety monitoring, health diagnosis, and reinforcement maintenance (Zhang, Zhou, Han, Lin, & Wu, 2018).

In Brazil, different agencies at the Federal, State, and Municipal levels of government (public sector) manage bridges. The main bridge management system in Brazil is the *SGO - Sistema de Gerenciamento de Obras de Arte Especiais*, developed by the *DNIT - Departamento Nacional de Infraestruturae Transportes*. Around 137,000 bridges are estimated to exist in Brazil (Fausto Da Silva & Almeida De Melo, 2021). The DNIT oversees the design, construction, operation, maintenance, repair, rehabilitation, and replacement of more than 8000 bridges, tunnels, viaducts, walkways, and containment structures across the entire Brazilian road network (DNIT, 2020). In 2016, DNIT implemented the *PROARTE - Programa de Manutenção e Reabilitação de Estruturas*, which promotes and manages maintenance and rehabilitation services for these structures. So far, DNIT has

invested more than BRL 97 million in maintenance work on 505 existing infrastructures (DNIT, 2020).

2.2. Building information modeling in a bridge management system

A Bridge Management System (BMS) is an integrated bridge computer system designed to coordinate and control the entire process involving the bridge structure by providing decision support during the design, construction, operation, and maintenance phases. This system aims to ensure that limited resources are used reasonably to provide services to users in the best possible way and maintain operational bridges (Hurt & Schrock, 2016; Wan et al., 2019; Yin et al., 2011).

Implementing a BMS based on periodic bridge inspections aims to identify bridges requiring intervention actions such as maintenance, repair, or rehabilitation (Alonso Medina, León González, & Todisco, 2022; Alonso Medina & León González, 2022). From the information collected during the inspections, a database can be used to help understand the deterioration process of bridges under certain conditions to assist in planning maintenance actions and repair works (Alonso Medina et al., 2022; Alonso Medina & León González, 2022; Frangopol, Dong, & Sabatino, 2017). Decisions on tasks related to maintenance plans are usually made based on various historical data and documents accumulated throughout the construction life cycle. However, most of this data is in written media of different natures and data structures, complicating the interaction between the different infrastructure components (Chen & Wang, 2009; Motamedi, Hammad, & Asen, 2014).

Planned maintenance using BIM provides a centralized system that can enhance the visualization and information retrieval to quality maintenance and structure performance and reduce ongoing life cycle costs (Byun et al., 2021; Chan et al., 2016; Chong, Wang, Shou, Wang, & Guo, 2014, Chong et al., 2016). BIM application studies of the maintenance phase have been conducted in parallel with the design/build stages. Using BIM through a stakeholder collaboration platform aims to facilitate the development of high-complexity construction and maintenance projects (Ghaffarianhoseini et al., 2017; Wan et al., 2019). Implementing BIM in bridge maintenance management requires linking the information to a 3D model, and each piece of information must be appropriately distributed according to the information system (Byun et al., 2021). Topics on BIM application studies for bridge maintenance include safety diagnostics, schedule management, visualization of historical information, maintenance decision-making, and efficient asset management (Byun et al., 2021).

The information workflow in a BMS encompasses all structured and unstructured bridge data from the design, construction, operation, and maintenance phases. This data must be standardized and digitized through BIM for access by those interested in the bridge management process. Various researchers have made efforts to define standardized data delivery protocols for information management with BIM workflows (Bose, East, & Issa, 2021; Kim et al., 2018; Li, Zhang, Lin, Zhang, & Han, 2023; Moretti, Xie, Merino, Brazauskas, & Parlikad, 2020; Patacas, Dawood, & Kassem, 2020; Schwabe, Dichtl, König, & Koch, 2018; Thabet, Lucas, & Johnston, 2016; Tsay, Staub-French, & Poirier, 2022).

The COBie (Construction Operations Building Information Exchange) is an international standard published by the US Army Corps of Engineers in 2007 (East, 2007), which originated from the concept of Computer Aided Facility Management (CAFM) (Schwabe et al., 2018; Tsay et al., 2022). This specification describes processes and information requirements that simplify transferring specific data from the design and construction phases to the operation and maintenance of facilities. COBie proposes that data, documented in an article or in extensive electronic form, be collected and stored incrementally and systematically in digital format from the start of the project, allowing for a more efficient transition from the construction phase to the facility management phase, saving time and resources (Pärn & Edwards, 2017; Schwabe et al., 2018).

Moretti et al. (2020) and Patacas et al. (2020) focused on the open standards of BIM and created a framework for data integration. Moretti et al. (2020) developed an openBIM methodology to support asset management applications with limited availability of as-built information. For this purpose, they used an IFC entity for processing the geometric and semantic information of existing and newly created objects, supporting real-time data integration. As stated by Patacas et al. (2020), there is a challenge in effectively addressing the three capabilities of information management and maintenance management, as follows: (1) disseminating models and information, specifically Asset Information Models (AIM), sourced from multiple locations; (2) ensuring the validation of these information models against predefined requirements; and (3) utilizing the gathered information for effective facilities management, encompassing operational and maintenance activities. Thus, they developed a prototype Common Data Environment (CDE) based on using open standards and existing technologies to address these three issues.

Studies conducted by Kim et al. (2018) propose an innovative approach to efficiently manage BIM-based Facility Management (FM) information. The key idea was to establish a connection between IFC elements and FM. This connection was facilitated through the use of Semantic Web technology. Li et al. (2023) develop and demonstrate the utility of a BMS that utilizes BIM for managing bridge defects and maintenance. For standardization and unification of damage data during inspection, bridge defect information was classified and digitally coded according to the IFD (International Framework for Dictionaries) standard and used to create a database. The IFD standard was used to codify and transform the BIM data into a unified form for interactivity.

Poor management of structures has consistently been attributed to a lack of coordination and information during the maintenance process. Chong et al. (2014) studied improving maintenance quality and performance by

integrating BIM and facilities management. The results revealed that BIM creates a collaborative and effective vision within the facilities management platform, improving quality and performance. BIM implementation in the maintenance management of road bridges has been achieving similar results (Abudayyeh & Al-Battaineh, 2003; Chan et al., 2016; McGuire, Atadero, Clevenger, & Ozbek, 2016; Wan et al., 2019).

Abudayyeh and Al-Battaineh (2003) adopted a bridge as-built information model in their study to facilitate maintenance and rehabilitation decision-making by providing necessary data to assist management. As-built data provide historical information related to bridges' design, construction, operation, and maintenance, essential for planning maintenance and rehabilitation programs (Abudayyeh & Al-Battaineh, 2003). Koch et al. (2014) present the current achievements and open challenges in the automation of inspection processes. Although challenges remain under investigation, automated inspection methods are already a reality. Also, as-built 3D modeling, along with damage detection, has the potential to be fully automated to facilitate the management and maintenance of large structures (Koch et al., 2014).

McGuire et al. (2016) presented the implementation of a method and prototype to track and assess the structural condition of bridges. The search used commercial BIM software to link and analyze bridge inspection, evaluation, and management data. This research highlighted BIM's ability to maintain parametric information about the bridge and visually represent damage, which can be used in damage assessment and maintenance decision-making. The author's method allows the user to assess damage based on location and provides maintenance recommendations based on comparing current inspection results with previous inspections. Enhanced documentation of damage parameters during inspection processes with BIM capabilities can improve bridge management and maintenance with a more consistent dynamic process and deterioration data. However, for the future practical integration of BIM into bridge management, steps in software development need to be taken to adapt further BIM principles to the operation and maintenance of bridges' service lives.

Wan et al. (2019) developed a BMS based on BIM technology integrated with a geographic information system (GIS), which contains information on inspections and assessment of technical conditions to improve the efficiency of bridge maintenance management. In addition, IFC and international framework for dictionaries (IFD) standards are developed, implementing Web-BIM-oriented management with collaborative work among users. This Web-BIM-based BMS, coupled with IFC language development, predisposes more features to improve efficiency in managing and maintaining large structures. Other authors have also made advances in bridge management and maintenance using these features (Bazán, Alberti, Álvarez, & Trigueros, 2020; Byun et al., 2021; Tanaka, Hori, Onosato, Date, & Kanai, 2016).

3. Method

A comprehensive literature review was conducted in this work, aiming to identify the trends and challenges in recent and relevant studies in the area. Figure 3 shows a flowchart that summarizes the methodology adopted. Web of Science and Scopus Databases were adopted for bibliometric analysis since they are the main academic databases covering the study subjects. The bibliometric evaluations were carried out for the search parameter 'article title, abstract, keywords', starting with the terms 'bridge management' OR 'management of bridges'. From the results, the documents containing the keywords 'bridge maintenance' OR 'maintenance of bridges' were selected to subside the discussions.

The searches were performed in October 2023 and were restricted to journal articles and reviews published in English from 2013 to 2023. A careful review of the subject area and source titles was conducted to filter content out of the scope. The authors understand that the results do not fully represent the literature on Bridge management but bring a consistent sampling considering the database and criteria used.

4. Bridge management: scientific interest and trends

From the databases used and adopted criteria for the bibliometric analysis, 413 articles were identified addressing the central theme (bridge management). Figure 4 shows the occurrence by country (authors' institutions). The United States is the most influential and collaborative country, responding for 31.3% of the scientific production on the subject, followed by China (15.7%), the United Kingdom (5.8%), Italy (5.0%), South Korea (4.4%), Australia (4.0%), Canada (3.8%), and Portugal (3.4%).

As expected, large countries have large transportation networks, reflecting their interest and investment in bridge management efficiency and resulting in scientific production. Brazil, however, had until now a discrete international collaborative participation, with only four articles identified in the survey. Notably, 46 of the identified funding sponsors were administration offices or ministries linked to a country or state administration of transportation infrastructure. American Federal and State Offices were the most representatives in this aspect. This trend demonstrates the relevance and economic interest in efficiently managing transportation assets at a governmental level.

The interest in the subject is growing, as shown in Figure 5. After a stagnation between 2013 and 2016 (25 articles on average), significant growth in 2017, and stagnation between 2017 and 2019 (36 articles on average), the number of articles covering the subject consistently grew between 2020 and 2022, reaching 60 articles published in 2022 (2.5 times the number registered in 2013 and a growth of 23% compared to 2021). Fifty articles had already been published in 2023 by the date of completion of this research (October 2023).

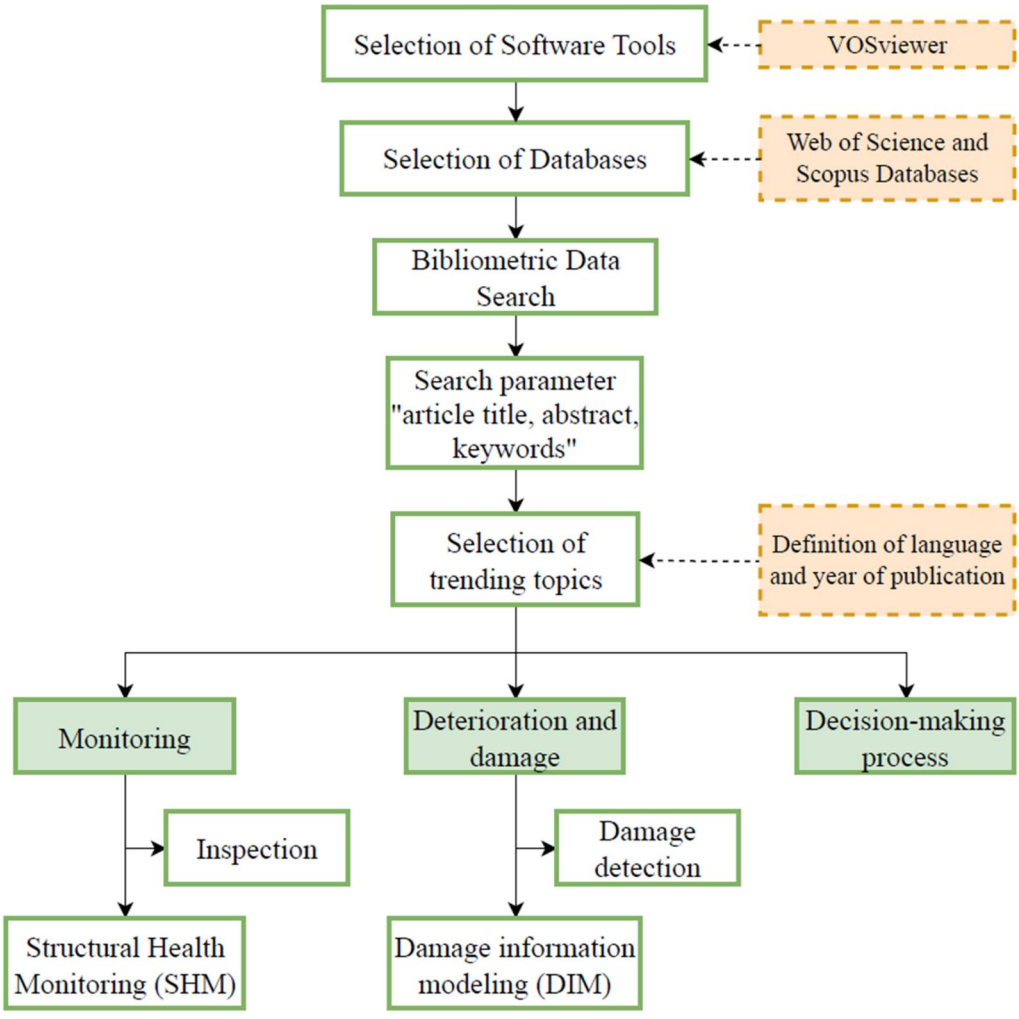


Figure 3. Overview of the search methodology and organization of the review.

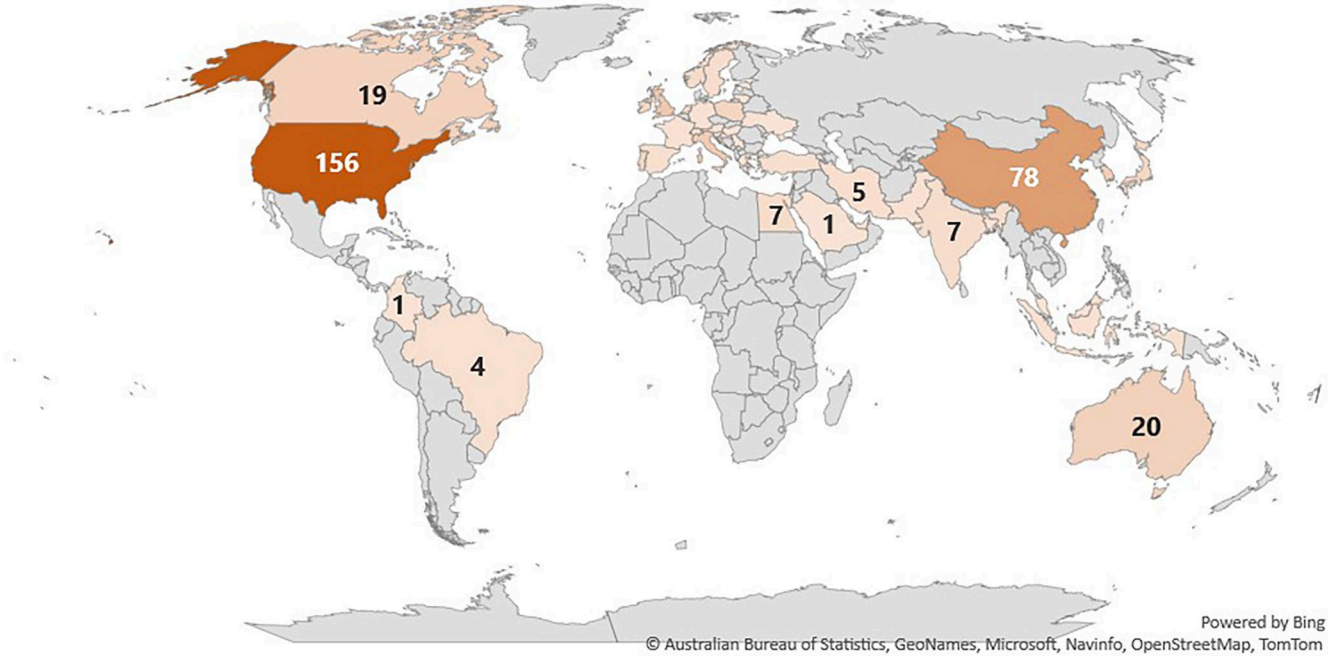


Figure 4. Countries by the number of published articles on the subject.

The evaluation shows that the strongest correlated keywords were optimization, deterioration, inspection, structural health monitoring, prediction, decision-making, information management, and bridge management systems. Figure 6 shows the overlay visualization of the co-occurrence network of the most used keywords weighted by occurrence and scored by publication year. The keywords were filtered, excluding the terms used in the search and generic ones. The analysis results show the trending topics in the bridge management context, summarized in Table 1. Table 2 lists the reviews considered in this work and summarizes the main subject, remarks, and contributions.

5. Literature review and remarks

5.1. Monitoring

Bridges are vital components of the road system, and to maintain adequate performance, functionality, and safety of structures throughout their service life, periodic inspection, maintenance, and rehabilitation actions must be implemented (Frangopol & Soliman, 2015; Xie et al., 2018). However, such structures are often located in hilly terrain that is difficult to access. Therefore, establishing effective monitoring, inspection methods, and a periodic maintenance strategy is paramount (Rashidi et al., 2020).



Figure 5. Annual publication trend on the subject from 2013 to 2023.

Condition monitoring of a bridge focuses primarily on determining in-service performance and detecting material degradation to assess the safety and service life of the structure (Abu Dabous & Feroz, 2020). Information sharing and data exchange keep pace with advances in bridge monitoring and management technologies. Moreover, they enable meaningful uses of information, service improvements, and enhanced bridge operation, maintenance, and safety (Delgado, Butler, Brilakis, Elshafie, & Middleton, 2018; Jeong et al., 2017).

5.1.1. Inspection

Bridge inspection information is the first step of the complete bridge management process and is the center of any BMS (Hurt & Schrock, 2016; Quirk, Matos, Murphy, & Pakrashi, 2018). The bridge inspector has as one of his main tasks the identification of damage (large or small) and potential problem areas in the structure before they become critical to structural or functional integrity (Bianchi, Sakib, Woolsey, & Hebdon, 2023). According to S. Xu et al. (2022), an inspection report contains information that can be divided into three categories: (1) basic inspection information (type of inspection, date of inspection, and personnel responsible); (2) information about individually identified damages; and (3) condition state classifications. Images taken on-site should be attached to improve damage

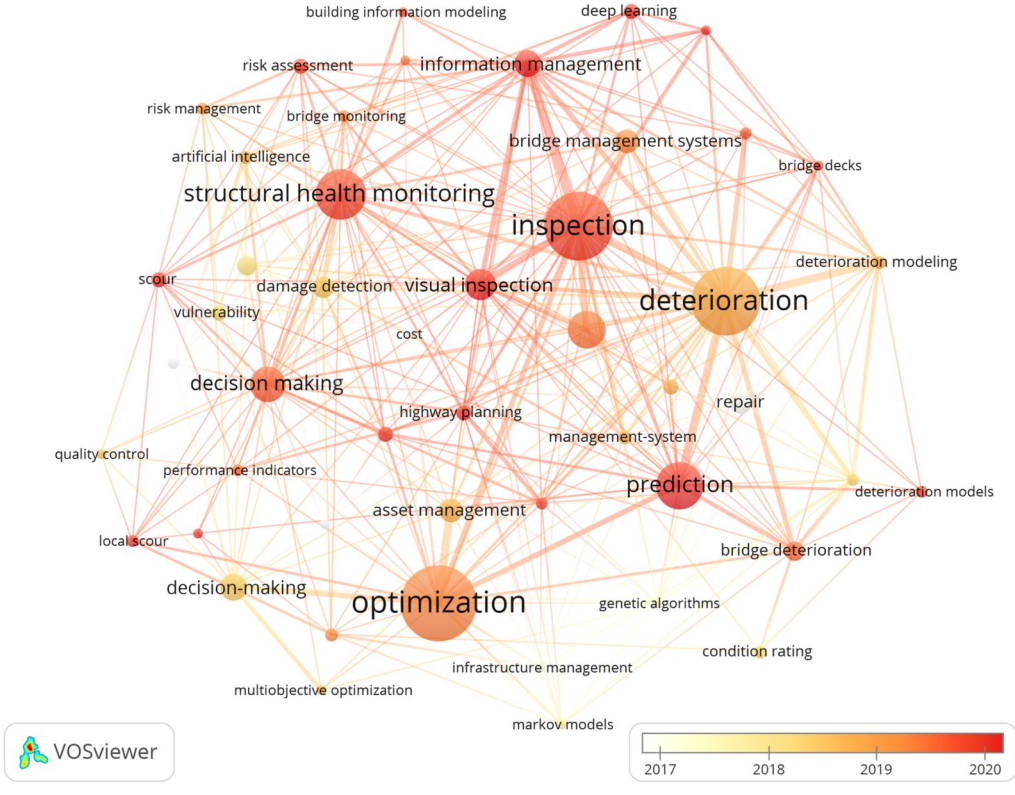


Figure 6. Overlay visualization of the co-occurrence network of the 60 most used keywords (excluding 'bridge management', 'bridge' and 'bridges').

Table 1. Trending topics related to the theme 'bridge management' according to the co-occurrence network obtained from the bibliometric analysis.

Trending topic	Some remarkable related topics	Cluster	Some recent references (5 years)
Machine Learning	Decision-making Damage detection	Structural Health Monitoring (SHM)	(Althaqafi & Chou, 2022; Asghari & Hsu, 2021; Karaaslan, Bagci, & Catbas, 2021; Mohammed Abdelkader, Marzouk, & Zayed, 2020; Saremi, Goulias, & Zhao, 2023; Wang, Ruan, & Xie, 2021; Xu & Guo, 2022; Yang, 2022)
Life Cycle	Asset management Decision-making	Optimization	(Akiyama, Frangopol, & Ishibashi, 2020; Allah Bukhsh, Stipanovic, Klanker, O' Connor, & Doree, 2019; Asghari & Hsu, 2021; Calvert, Neves, Andrews, & Hamer, 2020; Shim, Dang, Lon, & Jeon, 2019; Wang, Lee, & Gharaibeh, 2022; Xu & Guo, 2022)
Visual inspection	Prediction Bridge deterioration	Inspection	(Abu Dabous & Feroz, 2020; Alsharqawi, Zayed, & Abu Dabous, 2020; Dizaj, Padgett, & Kashani, 2021; Maroni, Tubaldi, McDonald, & Zonta, 2022; Santamaria Ariza, Zambon, S. Sousa, Campos e Matos, & Strauss, 2020; Turksezer, Iacovino, Giordano, & Limongelli, 2021)
Building Information Modeling (BIM)	Repair Condition assessment	Bridge Management System (BMS)	(Byun et al., 2021; Deng et al., 2021; Isailović et al., 2020; Kwon, Park, Park, & Lee, 2021; Mohammadi, Rashidi, Yu, & Samali, 2023; Wan et al., 2019)
Deep learning	Information management Deterioration Modeling	Deterioration	(Hong, Hwang, Chung, & Kim, 2022; Khandel & Soliman, 2021; Yang, 2022; Yue, Ding, Zhao, & Wang, 2022; Zhang et al., 2023)

description and subsequent quantitative measurements. The geometric information of the damage includes its shape characteristics (e.g. length, width, or area) and location (e.g. position relative to the bridge element and orientation of development).

Assessing the current condition of deteriorated structures is essential for identifying existing defect types in time and making appropriate repair strategies, and it becomes a challenging task that requires a large dataset and good expert knowledge and judgment (Bolar, Tesfamariam, & Sadiq, 2013; Panah & Kioumarsi, 2021; Xu & Turkan, 2019b). Rehabilitation, scheduling, and budget decisions are primarily based on the results of visual inspections of bridges, coupled with considerable uncertainty and subjectivity inherent in human judgments and manual data acquisition, which is labor-intensive and sometimes nearly impossible (Marzouk & Hisham, 2012; McGuire et al., 2016; Moufti, Zayed, & Dabous, 2014; Phares, Washer, Rolander, Graybeal, & Moore, 2004; Wojcik & Zarski, 2020). Visual inspection of bridges suffers from several limitations, including the safety of the inspection team, accuracy in detecting subsurface defects, and subjectivity of the process. Visual inspection is still the most cost-effective detection method; however, it is time-consuming, and the collected data is typically documented by completing standard inspection report forms (Abu Dabous, Yaghi, Alkass, & Moselhi, 2017; Agdas, Rice, Martinez, & Lasa, 2016; Al-Shalabi, Turkan, & Laflamme, 2015; McGuire et al., 2016; Valença, Dias-da-Costa, Gonçalves, Júlio, & Araújo, 2014). Digital bridge models are necessary to digitalize this process (Artus & Koch, 2022).

The use of Unmanned Aerial Vehicles (UAVs), mainly known as drones, makes inspection data acquisition faster and more reliable. These drones have changed the work routine, and there is great potential in automating this process. They can be equipped with different cameras, radars, laser scanners, infrared, thermal, and other types of sensors (Costin et al., 2018; Jensen, 2019; Wojcik & Zarski, 2020). UAVs have generated considerable interest in the architecture, engineering, construction, and facilities management (AEC-FM) sector due to their notable advantages, such as enhanced security, ease of use, mobility, and cost-effectiveness (Molina, Huang, & Jiang, 2023; Perry, Guo, Atadero, & van de Lindt, 2020; Xu & Turkan, 2019b). In addition, the use of Terrestrial Laser Scanner (TLS) is also a recent and accurate technology of great potential to be used in inspection processes. Laser scanners capture a large number of points for objects in a short period, forming a so-called 'point cloud' (M. Mohammadi et al., 2023; Panah & Kioumarsi, 2021; Rashidi et al., 2020; Sedek & Serwa, 2016; Volk, Stengel, & Schultmann, 2014; Xu & Turkan, 2019b).

The Ground Penetrating Radar (GPR) technique uses high-frequency electromagnetic waves to acquire subsurface information. It is a fast, non-destructive method implemented to inspect concrete bridges to detect the size and location of subsurface damage on bridges (Nguyen, Nguyen, Kang, Jang, & Shim, 2022; Prasanna et al., 2016; Shamsudin, Senin, Hamid, & Yusuf, 2015; Xu & Turkan, 2019b).

Table 2. List of recent reviews (5 years) found in the bibliometric analysis, with the main subject, limitations, and remarks.

References	Year	Main subject	Main remarks and contributions
(Jiménez Rios, Plevris, & Nogal, 2023)	2023	Bridge management using digital twin	Consensus on the adoption of digital twins for bridge design, management, and operation. Lack of software interoperability. Poor performance of anomaly-detection algorithms. Approach definition for integration at macro-scale. Implementation of industry 5.0 concepts.
(Saidin, Jamadin, Abdul Kudus, Mohd Amin, & Anuar, 2022)	2022	Vibration-Based Techniques	Applicability, pros, and cons of Ambient Vibrating Testing and Forced Vibrating Testing; Experimental Modal Analyses; and Benefits of the Stochastic Subspace Identification method.
(Niyirora et al., 2022)	2022	Vibration-based monitoring + machine learning	Vibration-based damage detection; Artificial intelligence and big data in damage detection
(Dayan, Chileshe, & Hassanli, 2022)	2022	Information modeling	Aspects of using information modeling in bridge management
(J. Yang et al., 2022)	2022	Big data knowledge engineering	Big data characteristics of bridge management
(Hosamo & Hosamo, 2022)	2022	Digital twin using 3D laser scanning	Digital twin and machine learning in BIM integration; laser scanning in bridge maintenance
(Bień & Salamak, 2022)	2022	Management of bridge structures	Classification of current and future generations of Bridge Management Systems
(Saback, Popescu, Blanksvärd, & Täljsten, 2022)	2022	Digital Twins and BIM	Several gaps are to be addressed. There is a lack of consensus on the definition of digital twins. Complex data flow and software compatibility.
(Fabianowski, Jakiel, & Stemplewski, 2021)	2021	Artificial neural network for condition assessment	Application potential of neural networks in bridge management and identification of difficulties as a guideline for further research
(Panah & Kioumarsi, 2021)	2021	BIM in health monitoring and maintenance process	A study on combining BIM with monitoring and maintenance of structures, presenting current limitations, and bringing solutions
(Ivanković, Strauss, & Sousa, 2020)	2020	Performance indicators for sustainable management	Performance indicators (PIs) for road bridges with Structural health monitoring (SHM)
(Sun, Shang, Xia, Bhowmick, & Nagarajaiah, 2020)	2020	SHM using Big Data (BD) and Artificial Intelligence (AI)	Perspectives and suggestions for using BD and AI in bridge SHM
(Srikanth & Arockiasamy, 2020)	2020	Deterioration models	Advantages and limitations of different bridge deterioration models
(Artus & Koch, 2020)	2020	Damage modeling	State of research and cases for use damage modeling, its types, categories, and frequency

Infrared (IR) thermography is another remote sensing technology whose basic theory is that the amount of heat conducted through a material will change in the presence of a subsurface defect, and it is used in the inspection and evaluation stages of bridges (Abu Dabous et al., 2017; Abu Dabous & Feroz, 2020; Xu & Turkan, 2019b). However, the ability of thermography to detect deep flaws depends on the depth and size of the defect, as well as factors in the environment where the test is being performed: the amount of sunlight, temperature differences, and test time (Xu & Turkan, 2019b, 2019a; Yehia, Abudayyeh, Nabulsi, & Abdelqader, 2007). A comparison between the inspection technologies listed in this section is summarized in Table 3.

Table 3. Comparison between inspection technologies.

Technology	Team safety	Access	Accuracy	Time spent	Cost	Skilled labour
UAVs	✓✓✓	✓✓✓	✓✓	✓	✓	✓✓
TLS	✓✓✓	✓✓	✓✓✓	✓✓✓	✓✓✓	✓✓✓
GPR	✓✓	✓	✓✓	✓✓	✓✓	✓✓✓
IR	✓✓✓	✓✓	✓✓	✓	✓	✓

✓ Low
✓✓ Medium
✓✓✓ High

The state of the bridge is the central point for the operating phase. Registering the state of conservation means registering the damages and defects of a structure (Artus & Koch, 2020). McGuire et al. (2016) developed a BIM-based method in which bridge inspectors capture information about the type of damage found, the amount, severity, and location. The location of the deterioration is essential to assess the bridge load and structure integrity and perform the intervention process (Adhikari, Moselhi, & Bagchi, 2013; Chan et al., 2016; Estes & Frangopol, 2003; Hüthwohl,

Brilakis, Borrmann, & Sacks, 2018; Prasanna et al., 2016). The lack of a description or documentation of the damaged site makes comparing results across inspections a limited task. Enhanced documentation of damage parameters with BIM during inspections can improve bridge management and link inspection results to analysis procedures (McGuire et al., 2016).

Al-Shalabi et al. (2015) implemented a 3D BrIM (Bridge Information Modeling) condition inspection framework with mobile device applications and cloud computing for inspection data documentation. In this approach, the information collected from the bridge elements can be accessed by on-site mobile devices such as a tablet computer that allows inspectors to access an accurate and up-to-date 3D information model of the bridge through cloud data storage services. The data cloud can be accessed from a home office and the site, enabling data sharing with all stakeholders simultaneously. This 3D model contains all bridge elements, including its object-level maintenance history. This BrIM-based inspection workflow allows input data such as damage type and dimensions to the database.

Xu and Turkan (2019b) developed a framework to create a BrIM with documented damage using unmanned aerial systems (UASs) to assist management and inspection practice. Images and videos collected more safely and quickly with UASs can be used to automatically detect cracks or other defects with the help of computer vision algorithms. Utilizing BrIM facilitates the allocation of defect information to specific model elements and allows for centralized management of all bridge-related data within a single model throughout the entire bridge's lifecycle (Xu & Turkan, 2019b).

Mohammadi et al. (2023) conducted a study that includes the use of TLS-derived BrIM for creating a detailed digital representation of a bridge, integrating geometrical and non-geometrical information of bridge elements. The study's methodology incorporates a condition assessment model for prioritizing bridge elements based on their health condition and integrates a Decision Support System (DSS) for evaluating remedial strategies and optimizing budget allocation and planning. The research article validates this methodology through a real case study and demonstrates its effectiveness in improving bridge management, safety, and maintenance.

5.1.2. Structural health monitoring

New monitoring technologies and practices are essential to circumvent visual inspection limitations and provide satisfactory results through secure and highly automated analysis procedures (Abu Dabous et al., 2017; Abu Dabous & Feroz, 2020; Agdas et al., 2016; Valença et al., 2014). SHM has played an efficient role in the operation and maintenance phase of structural life cycle management due to the high number of aging infrastructures (Panah & Kioumarsi, 2021; Theiler, Dragos, & Smarsly, 2018). SHM encompasses a range of methods and practices designed to assess the condition of a structure based on a combination of measurement, modeling, and analysis to obtain accurate information about them (Agdas et al., 2016; Panah & Kioumarsi, 2021; Pipinato, 2016).

The advances in SHM technology allied to BIM improve the quality of the assessment stages of structure management and decision-making (Boddupalli et al., 2019; Delgado et al., 2017, 2018; Panah & Kioumarsi, 2021; Rashidi et al., 2020). Real-time information collection using wireless sensor networks (WSN) and BIM technologies aims to improve accuracy in monitoring structures to reduce maintenance costs and improve accuracy in decision-making based on data collection and processing (Panah & Kioumarsi, 2021). SHM has been widely investigated and applied to large structures such as bridges (Comisu, Taranu, Boaca, & Scutaru, 2017; Gatti, 2019; Rashidi et al., 2020; Vardanega, Webb, Fidler, & Middleton, 2016; Worden & Cross, 2018).

SHM technology is applied to detect bridge quality problems, the need for intervention, identify regions with insufficient load capacity, the need for improvements, and assist in deciding in situations with high-risk factors for reconstruction or strengthening (Agdas et al., 2016; Comisu et al., 2017; Huston, Burns, & Razinger, 2016; Orcesi & Frangopol, 2011; Zhou et al., 2020). Surface defects such as cracking, spalling, and efflorescence are indicators of possible subsurface defects (such as reinforcement corrosion and concrete delamination), which are not visible but can directly reduce the structural capacity of the elements and be detrimental to the entire structure and need to be monitored regularly (Xu & Turkan, 2019b, 2019a).

Developing sensors and data acquisition technologies have enabled the installation of extensive monitoring systems on various structures (Comisu et al., 2017; Panah & Kioumarsi, 2021; Vardanega et al., 2016). SHM has been perceived as an expensive resource, performed only in large structures with specialized professionals. Making the structural monitoring system popular in small bridges and easy to apply has been addressed in several studies using Internet of Things (IoT) based real-time wireless sensors and mobile communication technology (Feldbusch, Sadegh-Azar, & Agne, 2017; Guzman-Acevedo et al., 2019; Sitton, Rajan, & Story, 2020; Yu et al., 2015).

The specific components of monitoring and maintenance are application-dependent and can vary significantly; however, most systems have the same fundamental elements, which are: (i) measurements by sensors and instrumentation; (ii) assessment of structural condition, and (iii) evaluation and prognosis to support decision-making related to maintenance and rehabilitation (Agdas et al., 2016; Alampalli & Ettouney, 2008; Huston et al., 2016). Automatically assessing structure integrity and locating damage can significantly reduce costs and improve safety in operations (Boddupalli et al., 2019). Table 4 summarises the studies that have used SHM, showing the objective of the research, the structure to which it was applied, its strategies, and the technologies used for monitoring.

Since monitoring is an integral part of the operation and maintenance phases, it is favorable to enable semantic descriptions of SHM systems using BIM to make monitoring information accessible, practical, and understandable

Table 4. Summary table of SHM strategies and technologies.

References	Aim	Structure application	Phase	Tools
(Yu et al., 2015)	Verify the feasibility of the mobile-SHM sensors and external sensors board	Concrete bridge with one tower and double cable plane	Operation	Smartphone sensors SHM
(Delgado et al., 2017)	Investigate the effect of concrete shrinkage and thermal strain transfer from the cast-in-situ deck and infill to the prestressed concrete girders	Prestressed concrete railway bridge	Construction	BIM + fibre-optic-based SHM
(Feldbusch et al., 2017)	Development of an 'iDynamics' app to study the differences between professional accelerometers and smartphone sensors	Pedestrian bridge	Operation	Smartphone sensors SHM + 'iDynamics' app
(Delgado et al., 2018)	Present a BIM approach to harness structural monitoring data dynamically with the automatic generation of parametric BIM models of systems	Steel-composite railway bridge	Construction	BIM + fibre-optic-based SHM
(Guzman-Acevedo et al., 2019)	Investigate the application of various sensors (GPS receivers, accelerometers, and smartphones) integrated with a smart sensor for the SHM of bridges	Reinforced concrete bridge	Operation	Smartphone sensors
(Boddupalli et al., 2019)	Enable the automated inventory of sensor data in the BIM environment to facilitate systematic maintenance and risk management	Reinforced concrete bridge	Operation	BIM + sensor data SHM
(Gatti, 2019)	Compare a static load test, required by Italian law, with a dynamic structural health monitoring (SHM) test, carried out at different times, to assess the structural reliability of a medium-length bridge built in the late 1960s.	Prestressed reinforced concrete	Operation	Measurement sensors
(Previtali, Brumana, & Banfi, 2022)	Test the feasibility of using portable mobile laser scanning for surveying large, complex infrastructures and compare it with TLS survey	Masonry old bridge	Operation	TLS + Mobile mapping systems (MMS) + photogrammetry
(Fawad et al., 2023)	Automating an existing bridge SHM system of a bridge by using BIM and finite element (FE) model	Concrete bridge	Operation	BIM + FE + sensors + IoT

(Delgado & Oyedele, 2020; Panah & Kioumarsi, 2021; Theiler et al., 2018). BIM includes tools, processes, and technologies for documenting and exchanging 3D models and can facilitate monitoring structures throughout the life cycle (Panah & Kioumarsi, 2021; Volk et al., 2014). The BIM database is represented by the IFC standard for the exchange phase, which allows the construction process to be standardized. However, not all monitoring information to describe and exchange construction information is supported by IFC (Ait-Lamallam, Sebari, Yaagoubi, & Doukari, 2021; Ait-Lamallam, Yaagoubi, Sebari, & Doukari, 2021; Theiler & Smarsly, 2018). Several studies propose extensions of the IFC schema to facilitate systems documentation and life cycle change management (Borrmann et al., 2015; Delgado & Oyedele, 2020; Delgado et al., 2018; Ji, Borrmann, Beetz, & Obergrießer, 2013; Rio, Ferreira, & Poças-Martins, 2013).

Combining the collection and analysis of data from SHM sensors with BIM capabilities has been examined in several studies (Boddupalli et al., 2019; Delgado & Oyedele, 2020; Delgado et al., 2017, 2018; Theiler et al., 2018; Zhou et al., 2020). Delgado et al. (2017) described a new approach that enables modeling sensor-based structural performance monitoring systems in a BIM environment. However, the interpretation of sensor data for decision-making purposes of long-term management strategies still

requires the skills of an experienced and competent professional.

Delgado et al. (2018) developed a dynamic BIM approach based on structural monitoring data to assist the decision-making process in a complementary study to the previously cited (Delgado et al., 2017). Parametric and semantically rich BIM models of structural monitoring systems were developed, allowing data visualization in a 3D environment. By incorporating BIM provisions during the operational phase of an asset, significant cost reductions can be provided by reducing tactile and visual inspections and maintenance. In more recent studies, Delgado et al. (2020) investigated current IFC open standard BIM data models to describe monitoring systems and circular economy precepts for constructed assets. However, they found that the IFC is not mature enough, limiting the interoperability and implementation of monitoring system modeling, data management, and analysis.

Boddupalli et al. (2019) used SHM coupled with BIM to improve the damage assessment of critical structures to facilitate maintenance and rehabilitation decision-making. The BIM visualization tool proposed in the study was implemented and validated using a real bridge, and SHM data were collected under varying temperature conditions. This approach integrated the sensor system and BIM to improve data management and visualization. The results showed the usefulness of the proposal in detecting and visualizing potential hazards during the life cycle of bridges, which benefits infrastructure owners in deciding on cost-effective future maintenance strategies.

5.2. Deterioration and damage

5.2.1. Damage detection

Deterioration is associated with operational factors, natural events, environmental effects, construction quality, age, geometry, and materials used. Each of these parameters can individually or jointly contribute differently to the failure of the structure (Bolar et al., 2013). Automatic damage detection methods have been implemented, especially for cracks, which are recurrent damage to concrete structures and can trigger failures (Adhikari et al., 2013; Prasanna et al., 2016; Torok, Golparvar-Fard, & Kochersberger, 2014; Y. Xu & Turkan, 2019b; Yu, Shen, & Shen, 2021; Zhang, Barri, Babanajad, & Alavi, 2020).

For crack analysis, identification and quantification of crack patterns are necessary to reveal the state of the bridges. Adhikari et al. (2013) proposed an integrated model based on digital image processing to detect and quantify cracks in concrete bridges, with 3D visualization that mimics local visual inspection. The proposed model comprised crack quantification, change detection, neural networks, and 3D visualization models. Furthermore, the authors noted that the numerical representation of concrete defects (quantification model) depends heavily on the camera resolution and imaging criteria, such as orthogonality.

Prasanna et al. (2016) presented a new automatic crack detection algorithm, the STRUM (spatially tuned robust multifeatured) classifier, combined with a robotic bridge scanning system, and as a result, 90% accuracy was obtained in thousands of tested cracks. In addition, the results can be quantified, archived, and compared over time. Zhang et al. (2020) also used a trained algorithm to detect cracks and fissures in concrete bridge decks, and the model accuracy was 99.25% in the tests. Yu et al. (2021) proposed a new YOLOv4-FPM model for real-time crack detection using UAVs or other fast-acquisition devices. The results show that the detection speed matches the real-time detection task, and the detection accuracy is sufficient to replace manual work.

In studies by Xu and Turkan (2019b), the defect information (type and severity) can be assigned to the 3D BrIM model, improving the visualization of the inspection data. This information is stored in a central object-oriented database in the IFC schema. However, several challenges and limitations associated with the implementation of the system were identified, primarily concerning the accuracy of crack detection, highly affected by the way images are collected and processed, and the difficulty with modeling the bridge geometry in 3D. The proposed framework did not offer support for automatic damage measurement, requiring manual data processing to build the BrIM and project the location of the damage on structural elements (Xu & Turkan, 2019b).

Valença et al. (2017) proposed a new method called MCrack-TLS to automatically assess cracks in concrete bridges based on combining image processing and terrestrial laser scanner (TLS) technology. TLS allows a precise orthorectification of images, solving one of the main disadvantages of applying image processing for crack characterization in large structures. The parameters evaluated were the width, length, orientation, and location of the cracks. These characteristics show that the proposed method can help define and optimize maintenance interventions on concrete bridges.

Wojcik and Zarski (2020) evaluated the applicability of modern methods of building bridges with BIM, 3D reconstruction, and Artificial Intelligence (AI). They concluded that the IFC scheme used in this way could describe a complex bridge geometry, damage information model, and geometry. Furthermore, it is possible to enrich this model with data captured by 3D reconstruction in the form of textured meshes. However, no currently available IFC viewer supports textured models, creating the need to develop and customize viewer apps in BIM.

Hüthwohlet al. (2019) proposed an objective classification of concrete damage, making it possible to classify various areas of potentially damaged bridges into their specific defect type following existing bridge inspection guidelines. Three separate pre-trained deep neural networks were tuned based on a multi-source dataset consisting of automatically collected image samples as well as various Department of Transportation inspection databases. In that study, they showed that this approach can reliably classify various types of defects with an average score of 85%. According to the authors, the limitations of the approach presented are related to training data. Some classes of defects were under-

represented compared to others. This is the main reason why the multiclassifier caused misclassification and bias (Hüthwohl et al., 2019). Isailović et al. (2020) described an IFC-based classification method for detecting and modeling fragmentation defects in BIM with more than 70% accuracy.

Based on the information about defects collected by the mobile terminal of the BMS developed by Wan et al. (2019), the evaluation module can perform automatic component analysis according to the maintenance pattern on the bridges. The BrIM displays different colors on bridge elements to intuitively describe the updated technical state at different levels. The green-colored elements remained in a safe condition, and the yellow-colored ones suffered slight damage. The red color means the elements have been severely damaged (Wan et al., 2019). Chan et al. (2016) used a similar modeling system, identifying the condition status attributed to elements with the colors green (fair condition), yellow (bad condition), and red (very bad condition), and included the white color to identify new elements or good condition.

Bridge repair and reinforcement information are stored in the system and can be consulted by users. When the defect develops to some extent and threatens the safety of the structure, the system automatically sends messages to bridge managers to remind them to create repair and reinforcement projects (Wan et al., 2019). The BIM-based system regularly generates statistics, graphs, and tables according to inspection and monitoring information, which help to make auxiliary maintenance decisions. After the reinforcement and repair procedure, it is possible to determine whether the maintenance is qualified or not, according to the actual situation. Maintenance can be viewed through the template, and managers can check repair results (Wan et al., 2019). Table 5 summarizes the damage detection studies mentioned.

With the help of equipment and the development of methods for damage detection and modeling, essential parameters are defined to optimize the maintenance stage. Locating the damage to the elements of the bridge structure is critical to improving inspection procedures and assisting in correct on-site intervention. Automatic damage detection methods represent significant advances in identifying and quantifying these data for bridge condition analysis. However, challenges and limitations are still associated with implementing these techniques and the total accuracy of the data collected and processed by such equipment.

5.2.2. Damage information modeling

Creating a damage information model (DIM) first requires the BIM of the structure since the damage is associated with the building element, its properties, and its material. BIM contains geometric and semantic construction information needed to model damage comprehensively, such as information about materials, actors, functions, and relations, assisting in planning new inspections, model simulations, maintenance planning, and execution of intervention services (Artus & Koch, 2020, 2022).

A DIM aims to address the issue of unsatisfactory interoperability resulting from the widespread use of BIM in operation and maintenance by providing a concept for the damaged data exchange between multiple processes. Standardizing damage information can help reduce information loss and cost (Artus & Koch, 2020, 2022; Hüthwohl et al., 2018). This standard needs to contain an information delivery manual (IDM), a model view definition (MVD), and a data format comparable to IFC, among others (Artus & Koch, 2020). An MVD is a computer implementation of an IDM (Sacks et al., 2018).

Studies from Sacks et al. (2018) bring a semantic enrichment method called SeeBridge, which introduces a novel approach to acquiring and compiling information for bridge inspection and systems management. The SeeBridge process utilizes an IDM to facilitate open data exchange, formally specifying user requirements and ensuring that the final model possesses semantic significance. Subsequently, based on the IDM, a Model View Definition (MVD) was prepared for the IFC4 Add2 data schema standard to facilitate the exchange of building information models.

The damage detection algorithm iterates over each BIM element and analyzes the images, form, and function in structure. Initially, images are used exclusively to locate visually detectable damage clusters. These findings undergo further refinement to identify specific damage types (structural crack, non-structural crack, fragmentation, flaking,

Table 5. Summary table of damage detection studies.

References	Component	Measurements	Tools	Resolution	Accuracy
(Prasanna et al., 2016)	Bridge Deck	Crack map: number of cracks per region	On-site robotic bridge scanning system + SCRUM algorithm	0.6 mm/px	90%
(Valença et al., 2017)	Concrete Bridge	Crack width, length, and orientation	TLS technology (MCrack-TLS) + image processing	0.35 mm/px	–
(Hüthwohl et al., 2019)	Concrete Bridge	Concrete bridge defects	UAV + textured models	0.1 mm/px	85%
(Xu & Turkan, 2019b)	Concrete Bridge	Detect cracks automatically	UAS + computer vision algorithms + BrIM	–	–
(Isailović et al., 2020)	Double girder concrete bridge	Spalling damage	Point-cloud + BIM + IFC schema	–	70%
(Zhang et al., 2020)	Concrete bridge decks	Crack detection	Neural network + long short-term memory	–	98.9%
(Wojcik & Zarski, 2020)	Prestressed box girder bridge	Crack	BIM + 3D reconstruction + AI + IFC schema	0.1 mm/px	97.5%

efflorescence, corrosion) using additional extracted properties such as element type, damage position, and damage location. Significant damage parameters (damage type, absolute, and relative size measurements) are extracted from the findings and incorporated into the BIM model (Sacks et al., 2018).

In a related study, Tanaka et al. (2016) proposed an information model for supporting the bridge inspection process based on the IFC and IFC-Bridge for BIM. Their work introduced entities that provide information about inspection tasks (measured regions and degradation elements), inspection results, and time variations of degradation were considered in the degradation elements. Also, the relation between the degradation and civil elements, and the civil elements and measured region were defined.

Hüthwohl et al. (2018) categorized in their study inspection information on reinforced concrete bridges using the IFC standard to integrate damage information into the BIM model. They listed different damage classes with their respective properties, such as diameter, depth, width, location, corrosion, stains, and cracks, with data taken from existing bridge inspection manuals. It can be noted that not all classes occur exclusively, several pathological manifestations occur together. It has been shown that the existing IFC4 standard can model bridge defects and general inspection information following existing bridge inspection guidelines. Limitations to this method exist in the extraction of defect properties, the general modeling approach, and the selection of IFC entities. Wan et al. (2019) also included parameters and attributed definitions to damage information, such as length, direction, and width.

Isailović et al. (2020) proposed an IFC semantic enrichment framework to inject the extracted and reconstructed damage features into the as-is IFC model. Semantic enrichment must meet two requirements: the damage characteristics must conform to the BMS damage classification, and the semantic data structure needs to conform to the IFC schema. The semantic information of damage includes damage type (classification of a visible surface defect), deterioration process (physical-chemical process causing surface defects), damage position (rough distance measure, relative to the dimension of the inspected element), damage extend (an approximate measure of the damage region), and damage severity (damage condition rating).

Studies from Artus and Koch (2022) focused on modeling damage information to support bridge review and structural analysis. A comprehensive DIM incorporating geometric, visual, and semantic damage data were studied and applied. Since so far, there are no damage-specific entities implemented in IFC, the authors have analyzed the available alternatives for modeling. A test of the IFC implementation concept was performed, and the alternatives were according to IFC 4 and a thorough visualization evaluation. Revit was used to design and export the beam model to IFC 4. The main results are the definition of the required data for damage, an object-oriented damage model that supports multiple use cases, and the implementation of the model in a standard. In addition, tests have shown that the standard is adequate for providing damaged information; however, a number of software does not adequately implement the standard.

In their research, Xu et al. (2022) introduced a novel approach to document and represent inspection-related information in bridge BIM models, focusing specifically on the geometric aspects of defects. The method involved modeling defects' spatial placement and shape representation parametrically. By leveraging the latest version of IFC, this work showcased a parametric-driven representation of defect information, contributing to the establishment of an integrated BIM environment for the lifecycle management of civil assets. The proposed IFC-based method offered three primary aspects of modeling within the BIM environment:

1. Inspection activity: This aspect focused on capturing and documenting the various inspection activities carried out on the bridge.
2. Individual defects: The method aimed to accurately model and represent the geometry and characteristics of individual defects found during inspections.
3. Logical relationships: The proposed approach also addressed defining and establishing logical relationships around the defects, providing a comprehensive understanding of their interconnections and implications.

By incorporating these elements into the BIM model, Xu et al. (2022) work presented an effective and systematic way to manage inspection-related information in bridge projects using BIM technology.

Semantic and geometric data are required to damage information processing, which includes measurements, such as crack width, length, orientation, and position of the element, and textual information, such as alphanumeric material descriptions (Artus & Koch, 2022). However, the integration of information related to inspection and specifically to damage within the BIM model of the bridge has not yet been fully accomplished. Current methods for describing the logical relationships between defects and other IFC entities are non-existent or incomplete (Hüthwohl et al., 2018; Isailović et al., 2020; Tanaka et al., 2016; Xu et al., 2022). There is no single way to model damage, so the input parameters differ from one study to another, as do the representation and material changes obtained. Specific entities for damage modeling have not yet been implemented in the IFC scheme, and an additional extension adds more complexity. Table 6 summarizes the main entities in the current IFC documentation for inspection and damage applied in some studies.

5.3. Decision-making process

The BMS has as one of its most important functions the decision-making process in bridge management, especially regarding the prioritization of maintenance, repair, rehabilitation, and replacement activities to optimize the restricted budgets (Contreras-Nieto, Shan, Lewis, & Hartell, 2019; Lee & Kim, 2007). Decision-making is critical to selecting which

Table 6. Summary table of the IFC entities for inspection and damage used in some of the studies cited.

IFC entity	Definition[1]	Application	References
IfcTask	Identifiable unit of work	Inspection process	(Hüthwohl et al., 2018; Isailović et al., 2020; Tanaka et al., 2016; S. Xu et al., 2022)
IfcElementAssembly	Represents complex element assemblies aggregated from several elements	Inspection finding	(Isailović et al., 2020; Sacks et al., 2018)
IfcSurfaceFeature	It is a modification on the surface of an element	Modeling defects that only affect the surface of a component	(Artus & Koch, 2022; Hüthwohl et al., 2018; Isailović et al., 2020; Sacks et al., 2018; G. Xu & Azhari, 2022)
IfcBuildingElementProxy	It is a proxy definition of an element, without a predefined meaning	Represent a defect	(Artus & Koch, 2022)
IfcAnnotation	It is an information element within the geometric context of a project	Add textual annotations about a defect to the component	(Artus & Koch, 2022)
IfcVoidingFeature	It is a modification of an element that reduces its volume	Modeling defects adding a void to a component	(Artus & Koch, 2022)
IfcImageTexture	It is a 2-dimensional texture that can be applied to the surface of a geometric item	Link the defect images on the surface of an object to the IFC model	(Hüthwohl et al., 2018; Sacks et al., 2018)

[1] *BuildingSMART International.*

bridges require maintenance and repair, and detailed information is necessary to select the most adequate and economical intervention strategies (Lee & Kim, 2007; Yehia, Abudayyeh, Fazal, & Randolph, 2008). There are considerable pressures to extend the lifetime of existing infrastructure, considering sustainability and its impact on the environment. This situation requires the industry to provide more durable and effective measures for protecting and repairing concrete structures (Matthews, 2007).

The decision-making process for repair and rehabilitation projects in bridges is influenced by several crucial factors. Yehia et al. (2008) identified some of these key considerations: (i) Nature, extent, and severity of the defect; (ii) Effect of the proposed repair method on the bridge's lifespan; (iii) Disruption to traffic flow during the repair process; and (iv) Availability of funds.

The first and most critical step in repairing a structure is correctly determining the cause of the damage. Knowing what caused the damage and reducing or eliminating that cause will repair last longer (von Fay, 2015). Like any reinforced concrete structure, infrastructure works suffer the loss of integrity over their lifetime due to degradation caused by chemical agents (alkali-aggregate reaction, corrosion, leaching, chloride and sulfate attacks, carbonation, efflorescence), physical (gradient of temperature, fatigue, overload, retraction, freeze-thaw cycles), biological (accumulation of organic matter, living organisms) or mechanical (DNIT, 2010; Helene, 2005; TxDOT, 2021; von Fay, 2015). Different deterioration processes generate different types of damage. Reinforcement corrosion is considered the predominant problem that causes several subsequent physical effects such as cracking, scaling, chipping, and delamination of concrete (Bolar et al., 2013; Cadenazzi, Dotelli, Rossini, Nolan, & Nanni, 2020; Moufti et al., 2014; Omar, Nehdi, & Zayed, 2017).

For every bridge that requires intervention, several strategies are available. These strategies range from 'doing nothing' to 'complete replacement', as not all damaged concrete requires an immediate repair unless the damage affects the safety or operating condition of the structure (Rashidi, 2013; von Fay, 2015). A decision tree is a helpful tool for classifying all possible alternatives and decision-making (Rashidi, 2013; Rashidi & Lemass, 2011). In many situations, the budget needed to carry out the intervention procedures is unavailable, and bridge managers must allocate the budget to higher-priority structures. In this case, 'doing nothing and monitoring' is a very common strategy that does not require investment (Rashidi & Lemass, 2011). This alternative is associated with the general monitoring of the condition of the elements, keeping them in service until an important action is needed, such as rehabilitation or replacement (Rashidi, 2013).

Yehia et al. (2008) proposed a decision support system for bridge maintenance. They suggest repair and rehabilitation strategies for common problems in reinforced concrete bridge decks, including the three most common issues, which are corrosion, delamination, and cracking. The authors developed a decision tree using knowledge obtained from the literature and research with experts. The decision tree starts with two main questions: the age of the bridge deck and the average daily traffic. Based on this last variable, the repair method can be managed differently. The next branches of the decision tree are the damages that will lead to a repair strategy. Once this strategy is selected, the tree moves to the level where it specifies the intervention method and ends with the repair material (Yehia et al., 2008).

Repair and minor rehabilitation lower the damage rate and restore bridge performance. Studies conducted by Bolar et al. (2013) proposed a hierarchical logic trial network to assess the condition of bridges. In this article, the authors demonstrated the difficulty of establishing a classification system for intervention actions for bridge elements. Parameters such as available materials and technologies, location, cost, urgency, and the current bridge condition, affect the selection criteria of repair techniques. It is also necessary to consider the different parameters to select the most appropriate repair technique (Mahdi, Khalil, Mahdi, & Mansour, 2019).

Mahdi et al. (2019) developed a decision support system for concrete bridges that assists in assessing the prioritization of the bridge maintenance strategy, helping to identify

the level of urgency for repair and selecting the most appropriate maintenance plan, considering budget constraints. A database was built, incorporating different factors into the assessment process, such as bridge defects, repair techniques, and associated costs.

Intervention selection involves a case-by-case assessment to determine the potential risks associated with any specific course of action. Bridge maintenance managers must analyze several criteria and consider constraints when proposing the best solution to rehabilitate a structure. Rashidi et al. (2010) developed a risk management framework for decision-support steps for bridge maintenance. This support system can provide appropriate intervention options based on cost, service life, product durability/sustainability, client preferences, and legal and environmental constraints.

Reinforcement corrosion is one of the most recurrent manifestations in concrete structures. Some studies bring intervention solutions for this specific damage to indicate repair to protect these severely deteriorated structures (S. Kim, Frangopol, & Zhu, 2011; Whitmore, 2018). A literature review allowed observing the lack of studies focused on intervention processes and procedures for reinforced concrete bridges. This information is commonly found in books, maintenance, and repair manuals for concrete structures and concrete bridges, with specific solutions for each type of damage (DNIT, 2010; P. Helene, 2005; P. R. Helene, 1992; JICA, 2014, 2017; NYSDOT, 2008; PennDOT, 2012; Souza & Ripper, 1998; TxDOT, 2021; VicRoads, 2018; von Fay, 2015; Whitmore, 2018).

5.4. Challenges and limitations

The constant expansion of transportation infrastructure and the aging and deterioration of existing infrastructure represent considerable and persistent challenges that require efficient and appropriate management. The high operational costs associated with BIM implementation (Lu, Fung, Peng, Liang, & Rowlinson, 2014) can be compensated by reducing costs related to bridge assessment, maintenance operations, and condition inspection (Al-Shalabi et al., 2015). BIM is planned to solve problems such as communication, integration, collaboration, and information interoperability (Al-Shalabi et al., 2015; Bradley, Li, Lark, & Dunn, 2016; Jiang, Jiang, Han, Wu, & Wang, 2019; Lu et al., 2014; Tawelian & Mickovski, 2016). However, the effective implementation of BIM in the BMS depends on tools and software that communicate efficiently with each other and the stakeholders (Floros & Ellul, 2021; Hajdin & Samec, 2022; Matarneh, Elghaish, Rahimian, Dawood, & Edwards, 2022).

There is a considerable interoperability problem between the different software due to the lack of a standardized neutral format for exchanging information (Ali, Chen, Srikonda, & Hu, 2014; Costin et al., 2018; Jiménez Rios et al., 2023). Between the same software company, there is usually a high level of data compatibility. However, interoperability is generally low between different companies, with a loss of information when exchanging data between software (Jiang et al., 2019). Interoperability is understood as the ability to exchange data between applications, facilitating workflows and process automation (Eastman, Teicholz, Sacks, & Liston, 2011).

OpenBIM is a collaborative project management process that uses BIM based on open standards and workflows, enhancing the benefits of BIM by ensuring the accessibility, reliability, management, and sustainability of digital data throughout the structure's life cycle (Ait-Lamallam, Yaagoubi, et al., 2021; Ciccone, Stasio, Asprone, Salzano, & Nicolella, 2022; Jiang et al., 2019; Moretti et al., 2020; Pour Rahimian, Chavdarova, Oliver, Chamo, & Potseluyko Amobi, 2019). OpenBIM standards have been developed by buildingSMART to promote interoperability, and mainly include five basic standards (IFC - Industry Foundation Classes, IDM—Information Delivery Manual, MVD—Model View Definitions, BCF—BIM Collaboration Format, and bsDD—buildingSMART Data Dictionary) (Jiang et al., 2019).

Most BIM creation software supports the import/export functionality of the IFC data schema, the main neutral data model developed by buildingSMART. The IFC schema is an extensible object-oriented data model divided into basic entities and sub-entities. This format classifies and structures data, allowing different sets of information to be easily extracted (Afsari, Eastman, & Castro-Lacouture, 2017; Pour Rahimian et al., 2019). The IFC schema is represented by logic-based EXPRESS language, and the semantic relationship is limited to the representation and sharing of inferred knowledge (K. Kim et al., 2018). The COBie standard promotes the use of open data formats, such as Extensible Markup Language (XML), SpreadsheetML, or the IFC STEP format, to guarantee interoperability between different systems and neutrality, making it easier to exchange information between the project's stakeholders (Bose et al., 2021; K. Kim et al., 2018; Schwabe et al., 2018).

Traditional inspection and management systems have proven inefficient for this new reality of the infrastructure network (Agdas et al., 2016; Shamsudin et al., 2015), and system modernization and automation are required (Boddupalli et al., 2019; Isailović et al., 2020). The inspections performed with the help of synchronized models in the data cloud can be accessed by on-site mobile devices and home office computer interfaces, making it possible for stakeholders to exchange data simultaneously (Al-Shalabi et al., 2015; Ding et al., 2016). In addition, SHM is also an expensive resource, so IoT is an alternative to enable the monitoring of small structures at lower costs (Y. Yu et al., 2015). However, one of the challenges of using mobile equipment on site is the availability of cell phone signals since most bridges are located in rural areas, far from the cities, and where cell phone service is unavailable (Al-Shalabi et al., 2015; Tsai, Hsieh, & Kang, 2014). To solve these problems, the researchers (Al-Shalabi et al., 2015; Tsai et al., 2014) suggest an offline approach to the BrIM system, allowing the inspector to download the files before arriving at the inspection site.

With advances in SHM technology and the automated inspection process combined with BIM, it is expected to improve the quality of the evaluation, management, damage

Table 7. Common challenges for damage detection and modeling in a BIM-based system.

Challenge	Solution	References
BIM implementation costs	Cost-benefit analysis	(Al-Shalabi et al., 2015; Bryde, Broquetas, & Volm, 2013; Lu et al., 2014; Tawelian & Mickovski, 2016)
Time required	Time-saving by optimization and collaborative work	(Al-Shalabi et al., 2015; Bryde et al., 2013; Lu et al., 2014)
Expensive resource	Internet of Things (IoT), wireless sensors, and mobile communication technology	(Ding et al., 2016; Feldbusch et al., 2017; Guzman-Acevedo et al., 2019; Sitton et al., 2020; Yu et al., 2015)
Visual inspections limitations	Monitoring technologies	(Abu Dabous et al., 2017; Abu Dabous & Feroz, 2020; Agdas et al., 2016; Isailović et al., 2020; McGuire et al., 2016; Shamsudin et al., 2015)
Low cell phone signals	Offline tool for inspection	(Al-Shalabi et al., 2015; Tsai et al., 2014)
Damage detection	Automated analysis procedures	(Adhikari et al., 2013; Boddupalli et al., 2019; Prasanna et al., 2016; Tran, Huh, Mac, Kang, & Han, 2018; Xu & Turkan, 2019b, 2019a; Zhang et al., 2020)
IFC limitations	Specific entities on the IFC scheme	(Ait-Lamallam, Sebari, et al., 2021; Ait-Lamallam, Yaagoubi, et al., 2021; Artus & Koch, 2022; Hüthwohl et al., 2018; Isailović et al., 2020; Rio et al., 2013; Tanaka et al., 2016; Theiler & Smarsly, 2018)

detection, and decision-making stages. The semantics of information in the IFC schema is required to exchange data between the 3D models and the monitoring data throughout the life cycle of the structure. However, not all information associated with the monitoring step can be described and exchanged within the IFC schema since IFC open standard models are not mature enough.

Damage detection and modeling in a BIM-based system have been developed and support the decision-making process for interventions. The correct identification and location of defects found in the structure are of great importance for proper repair decisions. Many advances have been made in damage modeling through image capture equipment and the development of data processing and analysis methods. However, there are still challenges and limitations in the implementation of these techniques and the automatic and accurate detection of defects, in addition to insertion into BIM models. The common and recurring challenges mentioned in this section are summarised in Table 7.

Thus, one way to circumvent these limitations and challenges is to combine visual inspections with SHM capabilities and technologies. Using portable non-destructive testing equipment, tablets, and smartphones to monitor the progression of damage and failures identified during a visual inspection is a valuable strategy for bridge maintenance management. This approach allows continuous monitoring of the problems identified by the inspector, maintaining the safe use of the structure and providing the necessary time for planning the intervention processes.

6. Conclusions

This review aimed to identify and discuss emerging topics in bridge maintenance focusing on comprehensive and integrated BIM-based BMSs. Advances, perspectives, and challenges in implementing such methods and technologies have been raised and discussed. Some concluding remarks are listed below:

- Aging structures, coupled with the increasing emergence of damage and failure, have led to an increase in research on damage modeling, with the approach of SHM and automatic damage detection using technological resources and process automation.
- From the bibliometric analysis, it was possible to observe an increase in the number of articles in this area between the years 2020 and 2022, showing the importance of the topic, especially in countries with large transportation networks, reflecting the interest in research and investment to improve the bridge management system.
- Visual inspection of bridges has limitations ranging from the subjectivity of the analysis to the inspection team's safety. Automatically assessing structure integrity through new monitoring technologies and practices enables more accurate, secure, and highly automated detection of damage and failures, significantly reducing operating costs and improving bridge safety. It does not mean substituting human inspection since the responsibility of validating the information cannot be delegated. However, it points to optimization with inspection assisted by a larger set of tools, with more data that could be processed by algorithms assisted by artificial intelligence.
- The implementation of automatic damage detection methods has been addressed, and it is observed that a large portion of the studies is directed mainly at cracking, which is recurring damage to concrete structures that cause subsequent damage.
- Semantic descriptions of inspections, SHM, and DIM using the IFC standard have been studied and implemented. As not all information is currently available in open standard data models, limiting the interoperability and implementation of these systems in BIM, several studies are investigating and proposing extensions to the IFC specifications.
- A BIM-based BMS represents centralized approaches that can improve information management and visualization processes, optimizing structure performance and reducing life cycle costs. Combined with SHM technologies, maintenance decision-making is accomplished by evaluating the state of the structure in real-time and monitoring any changes with the aid of sensors, laser scanners, and infrared thermography, among other technologies.
- Indeed, while the initial implementation of BIM may be perceived as a high-cost investment, its benefits can significantly outweigh these costs over the long term. BIM offers various advantages that can help offset expenses

and improve overall project efficiency. Some of the key benefits of BIM include (i) Cost reduction in assessment, maintenance, and inspections; (ii) Improved communication and collaboration; (iii) Enhanced integration of disciplines; (iv) Efficient project management; and (v) Data-driven decision making.

The current reality of bridge management is characterized by rapid and continuous changes. Within this context, various mature technologies and methodologies are already available. However, there remain significant challenges related to geographical coverage, data acquisition, transmission, computer processing, and analysis of large datasets. Moreover, the availability of equipment, trained personnel, and limitations in BIM data schema and model interoperability continue to pose barriers to implementation, particularly in the short term. It is also important to note that transportation authorities possess a substantial amount of data that requires consolidation in a BIM-based management system. In this regard, the use of Information Container for Linked Document Delivery (ISO Standard) is considered a pragmatic approach.

BMS tends to be multi-methodological, incorporating a diverse range of existing and emerging tools and methods. These include traditional human inspection techniques and highly automated SHM systems. BMSs must be highly flexible in processing data sets from different origins, complexity, and volume. Managing such diverse datasets is the main challenge to ensuring the compatibility and stratification of valuable and comprehensive information for safe and accurate decision-making. It involves sophisticated artificial intelligence, statistical and deterministic algorithms. Conversely, BIM is an essential part of the system, a data source, and a fundamental information repository. Standardization and well-defined protocols become indispensable to ensure interoperability, data sharing, and cost-effectiveness. These considerations extend to aspects such as the training and contracting of personnel, the production, and management of models, considering time and equipment constraints.

To the best of the authors' knowledge, there are currently no operational BIM-based BMS in practical use. However, ongoing initiatives by road authorities, such as the establishment of the Brazilian DNIT BIM nucleus driven by the Brazilian presidency's Act 10306 mandating BIM use in public administration engineering projects, signify progress. These efforts, in collaboration with academia and the private sector, are laying the groundwork for the creation of a new BMS that can adapt into a BIM-based system.

Acknowledgments

This study was financed by the Departamento Nacional de Infraestrutura de Transportes (DNIT), Brazil - TED 703/2020. The authors thank the support of DNIT and Universidade Federal de Viçosa and also acknowledge the collaboration of the research groups SICon and TechBIM.

Disclosure statement

No potential conflict of interest was reported by the authors.

References

Abu Dabous, S., & Feroz, S. (2020). Condition monitoring of bridges with non-contact testing technologies. *Automation in Construction*, *116*(October 2019), 103224. doi:10.1016/j.autcon.2020.103224

Abu Dabous, S., Yaghi, S., Alkass, S., & Moselhi, O. (2017). Concrete bridge deck condition assessment using IR Thermography and ground penetrating radar technologies. *Automation in Construction*, *81*(April), 340–354. doi:10.1016/j.autcon.2017.04.006

Abudayyeh, O., & Al-Battaineh, H. T. (2003). As-built information model for bridge maintenance. *Journal of Computing in Civil Engineering*, *17*(2), 105–112. doi:10.1061/(asce)0887-3801(2003)17:2(105)

Adhikari, R. S., Moselhi, O., & Bagchi, A. (2013). Image-based retrieval of concrete crack properties for bridge inspection. *Automation in Construction*, *39*, 180–194. doi:10.1016/j.autcon.2013.06.011

Afsari, K., Eastman, C. M., & Castro-Lacouture, D. (2017). JavaScript Object Notation (JSON) data serialization for IFC schema in web-based BIM data exchange. *Automation in Construction*, *77*, 24–51. doi:10.1016/j.autcon.2017.01.011

Agdas, D., Rice, J. A., Martinez, J. R., & Lasa, I. R. (2016). Comparison of visual inspection and structural-health monitoring as bridge condition assessment methods. *Journal of Performance of Constructed Facilities*, *30*(3), 1–10. doi:10.1061/(ASCE)CF

Ait-Lamallam, S., Sebari, I., Yaagoubi, R., & Doukari, O. (2021). IFCInfra4OM: An ontology to integrate operation and maintenance information in highway information modelling. *ISPRS International Journal of Geo-Information*, *10*(5), 305. doi:10.3390/ijgi10050305

Ait-Lamallam, S., Yaagoubi, R., Sebari, I., & Doukari, O. (2021). Extending the IFC standard to enable road operation and maintenance management through OpenBIM. *ISPRS International Journal of Geo-Information*, *10*(8), 496. doi:10.3390/ijgi10080496

Akcamete, A., Akinci, B., & Garrett, J. H. (2010). *Potential utilization of building information models for planning maintenance activities. Proceedings of the International Conference on Computing in Civil and Building Engineering, January*.

Akiyama, M., Frangopol, D. M., & Ishibashi, H. (2020). Toward life-cycle reliability-, risk- and resilience-based design and assessment of bridges and bridge networks under independent and interacting hazards: Emphasis on earthquake, tsunami and corrosion. *Structure and Infrastructure Engineering*, *16*(1), 26–50. doi:10.1080/15732479.2019.1604770

Alampalli, S., & Ettouney, M. (2008). Role of structural health monitoring in bridge security. *Bridge Structures*, *4*(3-4), 143–154. doi:10.1080/15732480802399417

Ali, N., Chen, S., Srikonda, R., & Hu, H. (2014). Development of concrete bridge data schema for interoperability. *Transportation Research Record: Journal of the Transportation Research Board*, *2406*(1), 87–97. doi:10.3141/2406-10

Allah Bukhsh, Z., Stipanovic, I., Klanker, G., O' Connor, A., & Doree, A. G. (2019). Network level bridges maintenance planning using multi-attribute utility theory. *Structure and Infrastructure Engineering*, *15*(7), 872–885. doi:10.1080/15732479.2017.1414858

Alonso Medina, P., León González, F. J., & Todisco, L. (2022). Data-driven prediction of long-term deterioration of RC bridges. *Construction and Building Materials*, *317*, 125790. doi:10.1016/j.conbuildmat.2021.125790

Alonso Medina, P., & León González, J. (2022). Reinforced concrete long-term deterioration prediction for the implementation of a Bridge Management System. *Materials Today: Proceedings*, *58*, 1265–1271. doi:10.1016/j.matpr.2022.02.033

Al-Shalabi, F. A., Turkan, Y., & Laflamme, S. (2015). BrIM implementation for documentation of bridge condition for inspection. 5th International/11th Construction Specialty Conference.

Alsharqawi, M., Zayed, T., & Abu Dabous, S. (2020). Integrated condition-based rating model for sustainable bridge management. *Journal*

of Performance of Constructed Facilities, *34*(5), 04020091. doi:10.1061/(asce)cf.1943-5509.0001490

Althaqafi, E., & Chou, E. (2022). Developing bridge deterioration models using an artificial neural network. *Infrastructures*, *7*(8), 101. doi:10.3390/infrastructures7080101

Artus, M., & Koch, C. (2020). State of the art in damage information modeling for RC bridges – A literature review. *Advanced Engineering Informatics*, *46*(November 2019), 101171. doi:10.1016/j.aei.2020.101171

Artus, M., & Koch, C. (2022). Object-oriented damage information modeling concepts and implementation for bridge inspection. *Journal of Computing in Civil Engineering*, *36*(6), 04022029. doi:10.1061/(ASCE)

American Society of Civil Engineers (ASCE). (2021). *2021 Infrastructure Report Card: Bridges*. https://infrastructurereportcard.org/cat-item/bridges/.

Asghari, V., & Hsu, S.-C. (2021). Upscaling complex project-level infrastructure intervention planning to network assets. *American Society of Civil Engineers*, *148*(1), 04021188. doi:10.1061/(ASCE)CO.1943

Bazán, Á. M., Alberti, M. G., Álvarez, A. A., & Trigueros, J. A. (2020). New perspectives for bim usage in transportation infrastructure projects. *Applied Sciences (Switzerland)*, *10*(20), 7072. doi:10.3390/app10207072

Benítez, P., Rocha, E., Varum, H., & Rodrigues, F. (2020). A dynamic multi-criteria decision-making model for the maintenance planning of reinforced concrete structures. *Journal of Building Engineering*, *27*(September 2018), 100971. doi:10.1016/j.jobe.2019.100971

Bianchi, E. L., Sakib, N., Woolsey, C., & Hebdon, M. (2023). Bridge inspection component registrationfor damage evolution. *Structural Health Monitoring*, *22*(1), 472–495. doi:10.1177/14759217221083647

Bień, J., & Salamak, M. (2022). The management of bridge structures – Challenges and possibilities. *Archives of Civil Engineering*, *68*(2), 5–35. 10.24425/ace.2022.140627.

Boddupalli, C., Sadhu, A., Rezazadeh Azar, E., & Pattyson, S. (2019). Improved visualization of infrastructure monitoring data using building information modeling. *Structure and Infrastructure Engineering*, *15*(9), 1247–1263. doi:10.1080/15732479.2019.1602150

Bolar, A., Tesfamariam, S., & Sadiq, R. (2013). Condition assessment for bridges: A hierarchical evidential reasoning (HER) framework. *Structure and Infrastructure Engineering*, *9*(7), 648–666. doi:10.1080/15732479.2011.602979

Borrmann, A., Kolbe, T. H., Donaubauer, A., Steuer, H., Jubierre, J. R., & Flurl, M. (2015). Multi-scale geometric-semantic modeling of shield tunnels for GIS and BIM applications. *Computer-Aided Civil and Infrastructure Engineering*, *30*(4), 263–281. doi:10.1111/mice.12090

Bose, D., East, E. W., & Issa, R. R. A. (2021). Impact of COBie on design activities. *Lecture Notes in Civil Engineering*, *98*, 672–682. doi:10.1007/978-3-030-51295-8_47

Bradley, A., Li, H., Lark, R., & Dunn, S. (2016). BIM for infrastructure: An overall review and constructor perspective. *Automation in Construction*, *71*, 139–152. doi:10.1016/j.autcon.2016.08.019

Bryde, D., Broquetas, M., & Volm, J. M. (2013). The project benefits of building information modelling (BIM). *International Journal of Project Management*, *31*(7), 971–980. doi:10.1016/j.ijproman.2012.12.001

buildingSMART. (2023). *Industry Foundation Classes (IFC)*. buildingSMART International. https://www.buildingsmart.org/standards/bsi-standards/industry-foundation-classes/.

Byun, N., Han, W. S., Kwon, Y. W., & Kang, Y. J. (2021). Development of BIM-based bridge maintenance system considering maintenance data schema and information system. *Sustainability (Switzerland)*, *13*(9), 4858. doi:10.3390/su13094858

Cadenazzi, T., Dotelli, G., Rossini, M., Nolan, S., & Nanni, A. (2020). Cost and environmental analyses of reinforcement alternatives for a concrete bridge. *Structure and Infrastructure Engineering*, *16*(4), 787–802. doi:10.1080/15732479.2019.1662066

Calvert, G., Neves, L., Andrews, J., & Hamer, M. (2020). Multi-defect modelling of bridge deterioration using truncated inspection records. *Reliability Engineering & System Safety*, *200*, 106962. doi:10.1016/j.ress.2020.106962

Chan, B., Guan, H., Hou, L., Jo, J., Blumenstein, M., & Wang, J. (2016). Defining a conceptual framework for the integration of modelling and advanced imaging for improving the reliability and efficiency of bridge assessments. *Journal of Civil Structural Health Monitoring*, *6*(4), 703–714. doi:10.1007/s13349-016-0191-6

Chen, H. M., & Wang, Y. H. (2009). *A 3-dimensional visualized approach for maintenance and management of facilities*. 2009 26th International Symposium on Automation and Robotics in Construction, ISARC 2009, Mm, 468–475. doi:10.22260/ISARC2009/0002

Chong, H. Y., Lopez, R., Wang, J., Wang, X., & Zhao, Z. (2016). Comparative analysis on the adoption and use of BIM in road infrastructure projects. *Journal of Management in Engineering*, *32*(6), 05016021. doi:10.1061/(asce)me.1943-5479.0000460

Chong, H. Y., Wang, J., Shou, W., Wang, X., & Guo, J. (2014). Improving quality and performance of facility management using building information modelling. Lecture Notes in Computer Science (Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics), 8683 LNCS, 44–50. doi:10.1007/978-3-319-10831-5_6

Ciccone, A., Stasio, S., Asprone, D., Salzano, A., & Nicolella, M. (2022). Application of openBIM for the management of existing railway infrastructure: Case study of the Cancello–Benevento Railway Line. *Sustainability (Switzerland)*, *14*(4), 2283. doi:10.3390/su14042283

Comisu, C. C., Taranu, N., Boaca, G., & Scutaru, M. C. (2017). Structural health monitoring system of bridges. *Procedia Engineering*, *199*, 2054–2059. doi:10.1016/j.proeng.2017.09.472

Contreras-Nieto, C., Shan, Y., Lewis, P., & Hartell, J. A. (2019). Bridge maintenance prioritization using analytic hierarchy process and fusion tables. *Automation in Construction*, *101*, 99–110. doi:10.1016/j.autcon.2019.01.016

Costin, A., Adibfar, A., Hu, H., & Chen, S. S. (2018). Building Information Modeling (BIM) for transportation infrastructure – Literature review, applications, challenges, and recommendations. *Automation in Construction*, *94*(October 2017), 257–281. doi:10.1016/j.autcon.2018.07.001

Dayan, V., Chileshe, N., & Hassanli, R. (2022). A scoping review of information-modeling development in bridge management systems. *Journal of Construction Engineering and Management*, *148*(9), 03122006. doi:10.1061/(asce)co.1943-7862.0002340

Delgado, J. M. D., Butler, L. J., Brilakis, I., Elshafie, M. Z. E. B., & Middleton, C. R. (2018). Structural performance monitoring using a dynamic data-driven BIM environment. *Journal of Computing in Civil Engineering*, *32*(3), 4018009. doi:10.1061/(asce)cp.1943-5487.0000749

Delgado, J. M. D., Butler, L. J., Gibbons, N., Brilakis, I., Elshafie, M. Z. E. B., & Middleton, C. (2017). Management of structural monitoring data of bridges using BIM. *Proceedings of the Institution of Civil Engineers - Bridge Engineering*, *170*(3), 204–218. doi:10.1680/jbren.16.00013

Delgado, J. M. D., & Oyedele, L. O. (2020). BIM data model requirements for asset monitoring and the circular economy. *Journal of Engineering, Design and Technology*, *18*(5), 1269–1285. doi:10.1108/JEDT-10-2019-0284

Deng, L., Lai, S., Ma, J., Lei, L., Zhong, M., Liao, L., & Zhou, Z. (2021). Visualization and monitoring information management of bridge structure health and safety early warning based on BIM. *Journal of Asian Architecture and Building Engineering*, *21*(2), 427–438. doi:10.1080/13467581.2020.1869013

Ding, Y., Han, R., Liu, H., Li, S., Zhao, X., & Yu, Y. (2016). Bridge inspection and management system based on smart phone. *Proceedings of the ASME 2016 Conference on Smart Materials, Adaptive Structures and Intelligent Systems*. http://www.asme.org/about-asme/terms-of-use. doi:10.1115/SMASIS2016-9067

Dizaj, E. A., Padgett, J. E., & Kashani, M. M. (2021). A Markov chain-based model for structural vulnerability assessmentof corrosion-damaged reinforced concrete bridges. *Philosophical Transactions. Series A, Mathematical, Physical, and Engineering Sciences*, *379*(2203), 20200290. doi:10.1098/rsta.2020.0290

DNIT. (2010). Manual de recuperação de pontes e viadutos rodoviários. Em *Departamento Nacional de Infraestrutura de Transportes*. Departamento Nacional de Infraestrutura de Transportes. http://ipr.dnit.gov.br/normas-e-manuais/manuais/documentos/744_manual_recuperacao_pontes_viadutos.pdf.

DNIT. (2020). *Com o PROARTE, DNIT já executou obras em 505 estruturas*. Departamento Nacional de Infraestrutura de Transportes. https://www.gov.br/dnit/pt-br/assuntos/noticias/com-o-proarte-dnit-ja-executou-obras-em-505-estruturas.

East, E. W. (2007). *Construction operations building information exchange (COBIE): Requirements definition and pilot implementation standard*. Champaign: U. S. Army, Corps of Engineers.

Eastman, C., Teicholz, P., Sacks, R., & Liston, K. (2011). *BIM Handbook: A guide to building information modeling for owners, managers, designers, engineers and contractors (2nd ed)*. New Jersey: John Wiley & Sons, Inc.

Estes, A. C., & Frangopol, D. M. (2003). Updating bridge reliability based on bridge management systems visual inspection results. *Journal of Bridge Engineering*, *8*(6), 374–382. doi:10.1061/(asce)1084-0702(2003)8:6(374)

Fabianowski, D., Jakiel, P., & Stemplewski, S. (2021). Development of artificial neural network for condition assessment of bridges based on hybrid decision making method – Feasibility study. Em. *Expert Systems with Applications*, *168*, 114271. doi:10.1016/j.eswa.2020.114271

Fanning, B., Clevenger, C. M., Ozbek, M. E., & Mahmoud, H. (2015). Implementing BIM on infrastructure: Comparison of two bridge construction projects. *Practice Periodical on Structural Design and Construction*, *20*(4), 4014044. doi:10.1061/(asce)sc.1943-5576.0000239

Fausto Da Silva, M. B. M., & Almeida De Melo, R. (2021). Condições de Pontes Rodoviárias: Cenário, Diagnóstico e Manutenção. Xii Cbpe 2021.

Fawad, M., Salamak, M., Poprawa, G., Koris, K., Jasinski, M., Lazinski, P., Lazinski, D., Piotrowski, M. Hasnain, Gerges, M. (2023). Automation of structural health monitoring (SHM) system of a bridge using BIMification approach and BIM-based finite element model development. *Scientific Reports*, *13*(1), 13215. doi:10.1038/s41598-023-40355-7

Feldbusch, A., Sadegh-Azar, H., & Agne, P. (2017). Vibration analysis using mobile devices (smartphones or tablets). *Procedia Engineering*, *199*, 2790–2795. doi:10.1016/j.proeng.2017.09.543

Flores-Colen, I., & De Brito, J. (2010). A systematic approach for maintenance budgeting of buildings faades based on predictive and preventive strategies. *Construction and Building Materials*, *24*(9), 1718–1729. doi:10.1016/j.conbuildmat.2010.02.017

Floros, G. S., & Ellul, C. (2021). Loss of information during design & construction for highways asset management: a geobim perspective. *ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, *VIII-4/W2-2021*(4/W2-2021), 83–90. doi:10.5194/isprs-annals-VIII-4-W2-2021-83-2021

Frangopol, D. M., Dong, Y., & Sabatino, S. (2017). Bridge life-cycle performance and cost: Analysis, prediction, optimisation and decision-making. *Structure and Infrastructure Engineering*, *13*(10), 1239–1257. doi:10.1080/15732479.2016.1267772

Frangopol, D. M., & Kim, S. (2011). Service life, reliability and maintenance of civil structures. Service Life Estimation and Extension of Civil Engineering Structures, 145–178. doi:10.1533/9780857090928.2.145

Frangopol, D. M., & Soliman, M. (2015). Life-cycle of structural systems: Recent achievements and future directions. *Structure and Infrastructure Engineering*, *12*(1), 1–20. doi:10.1080/15732479.2014.999794

Gatti, M. (2019). Structural health monitoring of an operational bridge: A case study. *Engineering Structures*, *195*(vember 2018), 200–209. doi:10.1016/j.engstruct.2019.05.102

Ghaffarianhoseini, A., Tookey, J., Ghaffarianhoseini, A., Naismith, N., Azhar, S., Efimova, O., & Raahemifar, K. (2017). Building Information Modelling (BIM) uptake: Clear benefits, understanding its implementation, risks and challenges. *Renewable and Sustainable Energy Reviews*, *75*(October 2015), 1046–1053. doi:10.1016/j.rser.2016.11.083

Gui, C., Zhang, J., Lei, J., Hou, Y., Zhang, Y., & Qian, Z. (2021). A comprehensive evaluation algorithm for project-level bridge maintenance decision-making. *Journal of Cleaner Production*, *289*, 125713. doi:10.1016/j.jclepro.2020.125713

Guzman-Acevedo, G. M., Vazquez-Becerra, G. E., Millan-Almaraz, J. R., Rodriguez-Lozoya, H. E., Reyes-Salazar, A., Gaxiola-Camacho, J. R., & Martinez-Felix, C. A. (2019). GPS, accelerometer, and smartphone fused smart sensor for SHM on real-scale bridges. *Advances in Civil Engineering*, *2019*, 1–15. doi:10.1155/2019/6429430

Hajdin, R., & Samec, V. (2022). BIM and BMS: Current status and challenges. *IABSE Symposium Prague, 2022: Challenges for existing and oncoming structures – Report*, 710–715. doi:10.2749/prague.2022.0710

Helene, P. (2005). *Manual de Reparo, Proteção e Reforço de Estruturas de Concreto*. São Paulo: Red Rehabilitar.

Helene, P. R. (1992). *Manual para reparo, reforço e proteção de estruturas de concreto (2a)*. São Paulo: Pini.

Hong, S. S., Hwang, C. H., Chung, S. W., & Kim, B. K. (2022). A deep-learning-based bridge damaged object automatic detection model using a bridge member model combination framework. *Applied Sciences (Switzerland)*, *12*(24), 12868. doi:10.3390/app122412868

Hosamo, H. H., & Hosamo, M. H. (2022). Digital twin technology for bridge maintenance using 3D laser scanning: A review. *Advances in Civil Engineering*, *2022*, 1–15. doi:10.1155/2022/2194949

Hurt, M., & Schrock, S. D. (2016). Bridge management. In. Highway bridge maintenance planning and scheduling, 289–310. Cambridge: Butterworth-Heinemann. doi:10.1002/9780470168059

Huston, D., Burns, D., & Razinger, J. (2016). *Structural health monitoring and maintenance aided by building information modelling and repair information tools. Proceedings of the 11th International Conference on Urban Regeneration and Sustainability (SC 2016), 1(Sc)*. 897–907. doi:10.2495/SC160731

Hüthwohl, P., Brilakis, I., Borrmann, A., & Sacks, R. (2018). Integrating RC bridge defect information into BIM models. *Journal of Computing in Civil Engineering*, *32*(3), 4018013. doi:10.1061/(asce)cp.1943-5487.0000744

Hüthwohl, P., Lu, R., & Brilakis, I. (2019). Multi-classifier for reinforced concrete bridge defects. *Automation in Construction*, *105*(April), 102824. doi:10.1016/j.autcon.2019.04.019

Isailović, D., Stojanovic, V., Trapp, M., Richter, R., Hajdin, R., & Döllner, J. (2020). Bridge damage: Detection, IFC-based semantic enrichment and visualization. *Automation in Construction*, *112*(December 2019), 103088. doi:10.1016/j.autcon.2020.103088

Isikdag, U., Underwood, J., & Aouad, G. (2008). An investigation into the applicability of building information models in geospatial environment in support of site selection and fire response management processes. *Advanced Engineering Informatics*, *22*(4), 504–519. doi:10.1016/j.aei.2008.06.001

ISO. (2018). Industry foundation classes (IFC) of data sharing in the construction and facility management industries. ISO, 16739-1.

Ivanković, A. M., Strauss, A., & Sousa, H. (2020). European review of performance indicators towards sustainable road bridge management. *Proceedings of the Institution of Civil Engineers: Engineering Sustainability*, *173*(3), 109–124. doi:10.1680/jensu.18.00052

Jensen, J. S. (2019). Innovative and sustainable operation and maintenance of bridges. *Structure and Infrastructure Engineering*, *16*(1), 72–83. doi:10.1080/15732479.2019.1604772

Jeong, S., Hou, R., Lynch, J. P., Sohn, H., & Law, K. H. (2017). An information modeling framework for bridge monitoring. *Advances in Engineering Software*, *114*, 11–31. doi:10.1016/j.advengsoft.2017.05.009

Jiang, S., Jiang, L., Han, Y., Wu, Z., & Wang, N. (2019). OpenBIM: An enabling solution for information interoperability. *Applied Sciences*, *9*(24), 5358. doi:10.3390/app9245358

JICA. (2014). *Bridge repair manual (Second)*. Department of Public Works and Highways. Tokyo: Japan International Cooperation Agency.

JICA. (2017). *Bridge repair manual*. Tokyo: Japan International Cooperation Agency.

Jiménez Rios, A., Plevris, V., & Nogal, M. (2023). Bridge management through digital twin-based anomaly detection systems: A systematic

review. Em. *Frontiers in Built Environment*, *9*, 1–18. doi:10.3389/fbuil.2023.1176621

Ji, Y., Borrmann, A., Beetz, J., & Obergrießer, M. (2013). Exchange of parametric bridge models using a neutral data format. *Journal of Computing in Civil Engineering*, *27*(6), 593–606. doi:10.1061/(asce)cp.1943-5487.0000286

Karaaslan, E., Bagci, U., & Catbas, N. (2021). A novel decision support system for long-term management of bridge networks. *Applied Sciences*, *11*(13), 5928. doi:10.3390/app11135928

Kendall, A., Keoleian, G. A., & Helfand, G. E. (2008). Integrated life-cycle assessment and life-cycle cost analysis model for concrete bridge deck applications. *Journal of Infrastructure Systems*, *14*(3), 214–222. doi:10.1061/(asce)1076-0342(2008)14:3(214)

Khandel, O., & Soliman, M. (2021). Integrated framework for assessment of time-variant flood fragility of bridges using deep learning neural networks. *Journal of Infrastructure Systems*, *27*(1), 04020045. doi:10.1061/(asce)is.1943-555x.0000587

Kim, K., Kim, H., Kim, W., Kim, C., Kim, J., & Yu, J. (2018). Integration of IFC objects and facility management work information using Semantic Web. *Automation in Construction*, *87*, 173–187. doi:10.1016/j.autcon.2017.12.019

Kim, S., Frangopol, D. M., & Zhu, B. (2011). Probabilistic optimum inspection/repair planning to extend lifetime of deteriorating structures. *Journal of Performance of Constructed Facilities*, *25*(6), 534–544. doi:10.1061/(asce)cf.1943-5509.0000197

Koch, C., Paal, S., Rashidi, A., Zhu, Z., König, M., & Brilakis, I. (2014). Achievements and challenges in machine vision-based inspection of large concrete structures. *Advances in Structural Engineering*, *17*(3), 303–318. doi:10.1260/1369-4332.17.3.303

Kwon, T. H., Park, S. H., Park, S. I., & Lee, S. H. (2021). Building information modeling-based bridge health monitoring for anomaly detection under complex loading conditions using artificial neural networks. *Journal of Civil Structural Health Monitoring*, *11*(5), 1301–1319. doi:10.1007/s13349-021-00508-6

Lee, C. K., & Kim, S. K. (2007). GA-based algorithm for selecting optimal repair and rehabilitation methods for reinforced concrete (RC) bridge decks. *Automation in Construction*, *16*(2), 153–164. doi:10.1016/j.autcon.2006.03.001

Li, S., Zhang, Z., Lin, D., Zhang, T., & Han, L. (2023). Development of a BIM-based bridge maintenance system (BMS) for managing defect data. *Scientific Reports*, *13*(1), 846. doi:10.1038/s41598-023-27924-6

Lu, W., Fung, A., Peng, Y., Liang, C., & Rowlinson, S. (2014). Cost-benefit analysis of Building Information Modeling implementation in building projects through demystification of time-effort distribution curves. *Building and Environment*, *82*, 317–327. doi:10.1016/j.buildenv.2014.08.030

Mahdi, I. M., Khalil, A. H., Mahdi, H. A., & Mansour, D. M. M. (2019). Decision support system for optimal bridge' maintenance. *International Journal of Construction Management*, *22*(3), 342–356. doi:10.1080/15623599.2019.1623991

Maroni, A., Tubaldi, E., McDonald, H., & Zonta, D. (2022). A monitoring-based classification system for risk management of bridge scour. *Proceedings of the Institution of Civil Engineers – Smart Infrastructure and Construction*, *175*(2), 92–102. doi:10.1680/jsmic.21.00016

Marzouk, M. M., & Hisham, M. (2012). *Bridge information modeling in sustainable bridge management*. ICSDC 2011: Integrating Sustainability Practices in the Construction Industry - Proceedings of the International Conference on Sustainable Design and Construction 2011, 457–466. doi:10.1061/41204(426)57

Matarneh, S., Elghaish, F., Rahimian, F. P., Dawood, N., & Edwards, D. (2022). Automated and interconnected facility management system: An open IFC cloud-based BIM solution. *Automation in Construction*, *143*, 104569. doi:10.1016/j.autcon.2022.104569

Matthews, S. (2007). CONREPNET: Performance-based approach to the remediation of reinforced concrete structures: Achieving durable repaired concrete structures. *Journal of Building Appraisal*, *3*(1), 6–20. doi:10.1057/palgrave.jba.2950063

McGuire, B., Atadero, R., Clevenger, C., & Ozbek, M. (2016). Bridge information modeling for inspection and evaluation. *Journal of Bridge Engineering*, *21*(4), 4015076. doi:10.1061/(asce)be.1943-5592.0000850

MLIT, (Ministry of Land, Infrastructure, Transport and Tourism, Japan). (2019). White Paper on Land, Infrastructure, Transport and Tourism in Japan, 2019.

Mohammadi, J., Guralnick, S. A., & Yan, L. (1995). Incorporating life-cycle costs in highway-bridge planning and design. *Journal of Transportation Engineering*, *121*(5), 417–424. doi:10.1061/(ASCE)0733-947X(1995)121:5(417)

Mohammadi, M., Rashidi, M., Yu, Y., & Samali, B. (2023). Integration of TLS-derived bridge information modeling (BrIM) with a decision support system (DSS) for digital twinning and asset management of bridge infrastructures. *Computers in Industry*, *147*, 103881. doi:10.1016/j.compind.2023.103881

Mohammed Abdelkader, E., Marzouk, M., & Zayed, T. (2020). A self-adaptive exhaustive search optimization-based method for restoration of bridge defects images. *International Journal of Machine Learning and Cybernetics*, *11*(8), 1659–1716. doi:10.1007/s13042-020-01066-x

Molina, A. A., Huang, Y., & Jiang, Y. (2023). A review of unmanned aerial vehicle applications in construction management: 2016–2021. *Standards*, *3*(2), 95–109. doi:10.3390/standards3020009

Moretti, N., Xie, X., Merino, J., Brazauskas, J., & Parlikad, A. K. (2020). An openbim approach to iot integration with incomplete as-built data. *Applied Sciences*, *10*(22), 8287. doi:10.3390/app10228287

Motamedi, A., Hammad, A., & Asen, Y. (2014). Knowledge-assisted BIM-based visual analytics for failure root cause detection in facilities management. *Automation in Construction*, *43*, 73–83. doi:10.1016/j.autcon.2014.03.012

Moufti, S., Zayed, T., & Dabous, S. (2014). Defect-based condition assessment of concrete bridges. *Transportation Research Record: Journal of the Transportation Research Board*, *2431*(1), 88–96. doi:10.3141/2431-12

Nguyen, D.-C., Kang, M., Jang, D., & Shim, C. (2022). Developing mixed reality-based digital twin model for bridge maintenance system. *The 22nd International Conference on Construction Applications of Virtual Reality*. https://www.researchgate.net/publication/365558657.

Nguyen, D.-C., Nguyen, T.-Q., Jin, R., Jeon, C.-H., & Shim, C.-S. (2022). BIM-based mixed-reality application for bridge inspection and maintenance. *Construction Innovation*, *22*(3), 487–503. doi:10.1108/CI-04-2021-0069

Nili, M. H., Taghaddos, H., & Zahraie, B. (2021). Integrating discrete event simulation and genetic algorithm optimization for bridge maintenance planning. *Automation in Construction*, *122*, 103513. doi:10.1016/j.autcon.2020.103513

Niyirora, R., Wei, J., Masengesho, E., Munyaneza, J., Niyonyungu, F., & Nyirandayisabye, R. (2022). Intelligent damage diagnosis in bridges using vibration-based monitoring approaches and machine learning: A systematic review. *Results in Engineering*, *16*, 100761. doi:10.1016/j.rineng.2022.100761

NYSDOT. (2008). *Fundamentals of bridge maintenance and inspection*. New York: New York State Department of Transportation.

Omar, T., Nehdi, M. L., & Zayed, T. (2017). Performance of NDT techniques in appraising condition of reinforced concrete bridge decks. *Journal of Performance of Constructed Facilities*, *31*(6), 4017104. doi:10.1061/(asce)cf.1943-5509.0001098

Orcesi, A. D., & Frangopol, D. M. (2011). Optimization of bridge maintenance strategies based on structural health monitoring information. *Structural Safety*, *33*(1), 26–41. doi:10.1016/j.strusafe.2010.05.002

Panah, R. S., & Kioumarsi, M. (2021). Application of building information modelling (BIM) in the health monitoring and maintenance process: A systematic review. *Sensors (Basel, Switzerland)*, *21*(3), 837. doi:10.3390/s21030837

Pärn, E. A., & Edwards, D. J. (2017). Conceptualising the FinDD API plug-in: A study of BIM-FM integration. *Automation in Construction*, *80*, 11–21. doi:10.1016/j.autcon.2017.03.015

Patacas, J., Dawood, N., & Kassem, M. (2020). BIM for facilities management: A framework and a common data environment using

open standards. *Automation in Construction*, *120*, 103366. doi:10.1016/j.autcon.2020.103366

PennDOT. (2012). Bridge Maintenance Manual. Pennsylvania Department of Transportation, *55*. ftp://ftp.dot.state.pa.us/public/PubsForms/Publications/PUB 55.pdf.

Perry, B. J., Guo, Y., Atadero, R., & van de Lindt, J. W. (2020). Streamlined bridge inspection system utilizing unmanned aerial vehicles (UAVs) and machine learning. *Measurement*, *164*, 108048. doi:10.1016/j.measurement.2020.108048

Phares, B. M., Washer, G. A., Rolander, D. D., Graybeal, B. A., & Moore, M. (2004). Routine highway bridge inspection condition documentation accuracy and reliability. *Journal of Bridge Engineering*, *9*(4), 403–413. doi:10.1061/(asce)1084-0702(2004)9:4(403)

Pipinato, A. (2016). Bridge assessment, retrofit, and management. In A. Pipinato (Org.), *Innovative bridge design handbook: construction, rehabilitation and maintenance* (p. 721–757). Boston: Butterworth-Heinemann. doi:10.1016/B978-0-12-800058-8.00028-1

Pour Rahimian, F., Chavdarova, V., Oliver, S., Chamo, F., & Potseluyko Amobi, L. (2019). OpenBIM-Tango integrated virtual showroom for offsite manufactured production of self-build housing. *Automation in Construction*, *102*, 1–16. doi:10.1016/j.autcon.2019.02.009

Prasanna, P., Dana, K. J., Gucunski, N., Basily, B. B., La, H. M., Lim, R. S., & Parvardeh, H. (2016). Automated crack detection on concrete bridges. *IEEE Transactions on Automation Science and Engineering*, *13*(2), 591–599. doi:10.1109/TASE.2014.2354314

Previtali, M., Brumana, R., & Banfi, F. (2022). Existing infrastructure cost effective informative modelling with multisource sensed data: TLS, MMS and photogrammetry. *Applied Geomatics*, *14*(S1), 21–40. doi:10.1007/s12518-020-00326-3

Quirk, L., Matos, J., Murphy, J., & Pakrashi, V. (2018). Visual inspection and bridge management. *Structure and Infrastructure Engineering*, *14*(3), 320–332. doi:10.1080/15732479.2017.1352000

Rashidi, M. (2013). Decision support system for remediation of concrete bridges. University of Wollongong Thesis Collections Department, 1954–2016.

Rashidi, M., & Lemass, B. (2011). A decision support methodology for remediation planning of concrete bridges. *Journal of Construction Engineering and Project Management*, *1*(2), 1–10. doi:10.6106/JCEPM.2011.1.2.001

Rashidi, M., Lemass, B., Gibson, P., Lu, J. W. Z., Leung, A. Y. T., Iu, V. P., & Mok, K. M. (2010). A decision support system for concrete bridge maintenance. *AIP Conference Proceedings*, *1233*(PART 1), 1372–1377. doi:10.1063/1.3452105

Rashidi, M., Mohammadi, M., Kivi, S. S., Abdolvand, M. M., Truong-Hong, L., & Samali, B. (2020). A decade of modern bridge monitoring using terrestrial laser scanning: Review and future directions. *Remote Sensing*, *12*(22), 3796. doi:10.3390/rs12223796

Ren, G., Ding, R., & Li, H. (2019). Building an ontological knowledge-base for bridge maintenance. *Advances in Engineering Software*, *130*(February), 24–40. doi:10.1016/j.advengsoft.2019.02.001

Rio, J., Ferreira, B., & Poças-Martins, J. (2013). Expansion of IFC model with structural sensors. *Informes de la Construcción*, *65*(530), 219–228. doi:10.3989/ic.12.043

Saback, V., Popescu, C., Blanksvärd, T., & Täljsten, B. (2022). Asset management of existing concrete bridges using digital twins and BIM: a state-of-the-art literature review. *Nordic Concrete Research*, *66*(1), 91–111. Index (ESCI) doi:10.2478/ncr-2021-0020

Sacks, R., Kedar, A., Borrmann, A., Ma, L., Brilakis, I., Hüthwohl, P., … Muhic, S. (2018). SeeBridge as next generation bridge inspection: Overview, information delivery manual and model view definition. *Automation in Construction*, *90*(May 2017), 134–145. doi:10.1016/j.autcon.2018.02.033

Saidin, S. S., Jamadin, A., Abdul Kudus, S., Mohd Amin, N., & Anuar, M. A. (2022). An overview: the application of vibration-based techniques in bridge structural health monitoring. *International Journal of Concrete Structures and Materials*, *16*(1), 1–17. doi:10.1186/s40069-022-00557-1

Sánchez-Silva, M., Frangopol, D. M., Padgett, J., & Soliman, M. (2016). Maintenance and operation of infrastructure systems: Review. *Journal of Structural Engineering*, *142*(9), 1–16. doi:10.1061/(asce)st.1943-541x.0001543

Santamaria Ariza, M., Zambon, I., S. Sousa, H., Campos e Matos, J. A., & Strauss, A. (2020). Comparison of forecasting models to predict concrete bridge decks performance. *Structural Concrete*, *21*(4), 1240–1253. doi:10.1002/suco.201900434

Saremi, S. G., Goulias, D., & Zhao, Y. (2023). Alternative sequence classification of neural networks for bridge deck condition rating. *Journal of Performance of Constructed Facilities*, *37*(4), 04023025. doi:10.1061/JPCFEV.CFENG-4390

Schwabe, K., Dichtl, M., König, M., & Koch, C. (2018). COBie – A specification for the Construction Operations Building Information Exchange. In A. Borrmann, M. König, C. Koch, & J. Beetz (Orgs.), *Building Information Modeling: Technology Foundations and Indrustry Practice*. Cham: Springer. doi:10.1007/978-3-319-92862-3_9

Sedek, M., & Serwa, A. (2016). Development of new system for detection of bridges construction defects using terrestrial laser remote sensing technology. *The Egyptian Journal of Remote Sensing and Space Science*, *19*(2), 273–283. doi:10.1016/j.ejrs.2015.12.005

Shamsudin, A. M., Senin, S. F., Hamid, R., & Yusuf, K. (2015). Concrete delaminations location and its severity detection by visual inspection and ground penetrating radar. *Journal of Engineering Science and Technology*, *10*, 1–12.

Shepard, R. W. (2005). Bridge management issues in a large agency. *Structure and Infrastructure Engineering*, *1*(2), 159–164. doi:10.1080/15732470412331289378

Shim, C. S., Dang, N. S., Lon, S., & Jeon, C. H. (2019). Development of a bridge maintenance system for prestressed concrete bridges using 3D digital twin model. *Structure and Infrastructure Engineering*, *15*(10), 1319–1332. doi:10.1080/15732479.2019.1620789

Sitton, J. D., Rajan, D., & Story, B. A. (2020). Bridge frequency estimation strategies using smartphones. *Journal of Civil Structural Health Monitoring*, *10*(3), 513–526. doi:10.1007/s13349-020-00399-z

Souza, V. C. M., & Ripper, T. (1998). Patologia, Recuperação E Reforço De Estruturas De Concreto. *Pini* (1a ed.). São Paulo: Pini.

Srikanth, I., & Arockiasamy, M. (2020). Deterioration models for prediction of remaining useful life of timber and concrete bridges: A review. *Journal of Traffic and Transportation Engineering (English Edition)*, *7*(2), 152–173. doi:10.1016/j.jtte.2019.09.005

Sun, L., Shang, Z., Xia, Y., Bhowmick, S., & Nagarajaiah, S. (2020). Review of bridge structural health monitoring aided by big data and artificial intelligence: From condition assessment to damage detection. *Journal of Structural Engineering*, *146*(5), 04020073. doi:10.1061/(asce)st.1943-541x.0002535

Tanaka, F., Hori, M., Onosato, M., Date, H., & Kanai, S. (2016). Bridge information model based on IFC standards and web content providing system for supporting an inspection process. *Proc. in International Conference on Computing in Civil and Building Engineering (ICCCBE)*, 1140–1147. http://www.see.eng.osaka-u.ac.jp/seeit/icccbe2016/Proceedings/Full_Papers/144-301.pdf.

Tawelian, L. R., & Mickovski, S. B. (2016). The implementation of geotechnical data into the BIM process. *Procedia Engineering*, *143*, 734–741. doi:10.1016/j.proeng.2016.06.115

Thabet, W., Lucas, J., & Johnston, S. (2016). A case study for improving BIM-FM handover for a large educational institution. *Construction Research Congress*, 2177–2186. doi:10.1061/9780784479827.217

Theiler, M., Dragos, K., & Smarsly, K. (2018). Semantic description of structural health monitoring algorithms using building information modeling. Proceedings of the Workshop of the European Group for Intelligent Computing in Engineering, 10864 LNCS, 150–170. doi:10.1007/978-3-319-91638-5_8

Theiler, M., & Smarsly, K. (2018). IFC Monitor – An IFC schema extension for modeling structural health monitoring systems. *Advanced Engineering Informatics*, *37*, 54–65. doi:10.1016/j.aei.2018.04.011

Torok, M. M., Golparvar-Fard, M., & Kochersberger, K. B. (2014). Image-based automated 3D crack detection for post-disaster

building assessment. *Journal of Computing in Civil Engineering*, *28*(5), 1–13. doi:10.1061/(asce)cp.1943-5487.0000334

Tran, Q. H., Huh, J., Mac, V. H., Kang, C., & Han, D. (2018). Effects of rebars on the detectability of subsurface defects in concrete bridges using square pulse thermography. *NDT & E International*, *100*, 92–100. doi:10.1016/j.ndteint.2018.09.001

Tsai, Y.-H., Hsieh, S.-H., & Kang, S.-C. (2014). A BIM-enabled approach for construction inspection. *Computing in Civil and Building Engineering*, 721–728. doi:10.1061/9780784413616.090

Tsay, G. S., Staub-French, S., & Poirier, É. (2022). BIM for facilities management: an investigation into the asset information delivery process and the associated challenges. *Applied Sciences*, *12*(19), 9542. doi:10.3390/app12199542

Turksezer, Z. I., Iacovino, C., Giordano, P. F., & Limongelli, M. P. (2021). Development and implementation of indicators to assess bridge inspection practices. *Journal of Construction Engineering and Management*, *147*(12), 04021165. doi:10.1061/(asce)co.1943-7862.0002195

TxDOT. (2021). *Concrete repair manual* (G. A. Bettis, Org.). Austin: Texas Department of Transportation.

Valença, J., Dias-da-Costa, D., Gonçalves, L., Júlio, E., & Araújo, H. (2014). Automatic concrete health monitoring: Assessment and monitoring of concrete surfaces. *Structure and Infrastructure Engineering*, *10*(12), 1547–1554. doi:10.1080/15732479.2013.835326

Valença, J., Puente, I., Júlio, E., González-Jorge, H., & Arias-Sánchez, P. (2017). Assessment of cracks on concrete bridges using image processing supported by laser scanning survey. *Construction and Building Materials*, *146*, 668–678. doi:10.1016/j.conbuildmat.2017.04.096

Vardanega, P. J., Webb, G. T., Fidler, P. R. A., & Middleton, C. R. (2016). Bridge monitoring. In A. Pipinato (Org.), *Innovative bridge design handbook: construction, rehabilitation and maintenance* (pp. 759–775). Boston: Butterworth-Heinemann. doi:10.1016/B978-0-12-800058-8.00029-3

VicRoads. (2018). Bridge maintenance and repair manual. *TAPPI Journal*, *17*(Número 7), 1–111. doi:10.32964/tj17.07

Volk, R., Stengel, J., & Schultmann, F. (2014). Building information modeling (BIM) for existing buildings – Literature review and future needs. *Automation in Construction*, *38*, 109–127. doi:10.1016/j.autcon.2013.10.023

von Fay, K. F. (2015). Guide to Concrete Repair (Second, Número August). U.S. Department of the Interior Bureau of Reclamation. doi:10.2307/j.ctt1x76d9j.11

Wan, C., Zhou, Z., Li, S., Ding, Y., Xu, Z., Yang, Z., … Yin, F. (2019). Development of a bridge management system based on the building information modeling technology. *Sustainability*,)*11*(17), 4583. doi:10.3390/su11174583

Wang, F., Lee, C.-C., & Gharaibeh, N. G. (2022). Network-level bridge deterioration prediction models that consider the effect of maintenance and rehabilitation. *Journal of Infrastructure Systems*, *28*(1), 05021009. doi:10.1061/(asce)is.1943-555x.0000662

Wang, K., Ruan, T., & Xie, F. (2021). LR-BCA: Label ranking for bridge condition assessment. *IEEE Access*. *9*, 4038–4048. doi:10.1109/ACCESS.2020.3048419

Whitmore, D. (2018). Galvanic cathodic protection of corroded reinforced concrete structures. *MATEC Web of Conferences*, *199*, 5006. doi:10.1051/matecconf/201819905006

Wojcik, B., & Zarski, M. (2020). Asesment of state-of-the-art methods for bridge inspection: Case study. *Archives of Civil Engineering*, *66*(4), 343–362. doi:10.24425/ace.2020.135225

Worden, K., & Cross, E. J. (2018). On switching response surface models, with applications to the structural health monitoring of bridges. *Mechanical Systems and Signal Processing*, *98*, 139–156. doi:10.1016/j.ymssp.2017.04.022

Wu, C., Wu, P., Wang, J., Jiang, R., Chen, M., & Wang, X. (2021). Ontological knowledge base for concrete bridge rehabilitation project management. *Automation in Construction*, *121*, 103428. doi:10.1016/j.autcon.2020.103428

Xie, H. B., Wu, W. J., & Wang, Y. F. (2018). Life-time reliability based optimization of bridge maintenance strategy considering LCA and LCC. *Journal of Cleaner Production*, *176*, 36–45. doi:10.1016/j.jclepro.2017.12.123

Xu, G., & Azhari, F. (2022). Data-driven optimization of repair schemes and inspection intervals for highway bridges. *Reliability Engineering & System Safety*, *228*, 108779. doi:10.1016/j.ress.2022.108779

Xu, G., & Guo, F. (2022). Sustainability-oriented maintenance management of highway bridge networks based on Q-learning. *Sustainable Cities and Society*, *81*, 103855. doi:10.1016/j.scs.2022.103855

Xu, S., Wang, J., Wang, X., Wu, P., Shou, W., & Liu, C. (2022). A parameter-driven method for modeling bridge defects through IFC. *Journal of Computing in Civil Engineering*, *36*(4), 04022015. doi:10.1061/(asce)cp.1943-5487.0001026

Xu, Y., & Turkan, Y. (2019a). Bridge inspection using bridge information modeling (BrIM) and unmanned aerialsystem (UAS). Advances in *informatics and computing in civil and construction engineer*ing, 617–624. Chicago: Springer International Publishing. doi:10.1007/978-3-030-00220-6

Xu, Y., & Turkan, Y. (2019b). BrIM and UAS for bridge inspections and management. *Engineering, Construction and Architectural Management*, *27*(3), 785–807. doi:10.1108/ECAM-12-2018-0556

Yang, D. Y. (2022). Deep reinforcement learning-enabled bridge management considering asset and network risks. doi:10.1061/(ASCE)IS.1943

Yang, J., Xiang, F., Li, R., Zhang, L., Yang, X., Jiang, S., Zhang, H., Wang, D., Liu, X. (2022). Intelligent bridge management via big data knowledge engineering. *Automation in Construction*, *135*, 104118. doi:10.1016/j.autcon.2021.104118

Yehia, S., Abudayyeh, O., Fazal, I., & Randolph, D. (2008). A decision support system for concrete bridge deck maintenance. *Advances in Engineering Software*, *39*(3), 202–210. doi:10.1016/j.advengsoft.2007.02.002

Yehia, S., Abudayyeh, O., Nabulsi, S., & Abdelqader, I. (2007). Detection of common defects in concrete bridge decks using nondestructive evaluation techniques. *Journal of Bridge Engineering*, *12*(2), 215–225. doi:10.1061/(asce)1084-0702(2007)12:2(215)

Yin, Z. H., Li, Y. F., Guo, J., & Li, Y. (2011). Integration research and design of the bridge maintenance management system. *Procedia Engineering*, *15*, 5429–5434. doi:10.1016/j.proeng.2011.08.1007

Yue, Z., Ding, Y., Zhao, H., & Wang, Z. (2022). Mechanics-guided optimization of an LSTM network for real-time modeling of temperature-induced deflection of a cable-stayed bridge. *Engineering Structures*, *252*, 113619. doi:10.1016/j.engstruct.2021.113619

Yu, Y., Han, R., Zhao, X., Mao, X., Hu, W., Jiao, D., Li, M., Ou, J. (2015). Initial validation of mobile-structural health monitoring method using smartphones. *International Journal of Distributed Sensor Networks*, *11*(2), 274391. doi:10.1155/2015/274391

Yu, Z., Shen, Y., & Shen, C. (2021). A real-time detection approach for bridge cracks based on YOLOv4-FPM. *Automation in Construction*, *122*(December 2020), 103514. doi:10.1016/j.autcon.2020.103514

Zhang, L., Zhou, G., Han, Y., Lin, H., & Wu, Y. (2018). Application of internet of things technology and convolutional neural network model in bridge crack detection. *IEEE Access*. *6*, 39442–39451. doi:10.1109/ACCESS.2018.2855144

Zhang, Q., Barri, K., Babanajad, S. K., & Alavi, A. H. (2020). Real-time detection of cracks on concrete bridge decks using deep learning in the frequency domain. *Engineering*, *7*(12), 1786–1796. doi:10.1016/j.eng.2020.07.026

Zhang, X., Eric Wogen, B., Chu, Z., Dyke, S. J., Poston, R., Hacker, T., Ramirez, J., Liu, X., Iturburu, L., Baah, P., Hunter, J. (2023). Machine-supported bridge inspection image documentation using artificial intelligence. *Transportation Research Record: Journal of the Transportation Research Board*, *2677*(5), 720–736. doi:10.1177/03611981221135803

Zhou, C., Wu, Y., Cui, G., Zhang, A., Gao, Y., Wang, X., Ouyang, J., Sun, H., Liang, Y., Liu, Z., Zhang, L. (2020). Comprehensive measurement techniques and multi-index correlative evaluation approach for structural health monitoring of highway bridges. *Measurement*, *152*, 107360. doi:10.1016/j.measurement.2019.107360